



Advances in computer vision for comprehensive railway engineering: from track inspection to rolling stock and safety monitoring

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Abstract

Railways form a vital part of global transportation networks, supporting economic development and sustainability through the efficient movement of freight and passengers. Maintaining the safety and reliability of these systems requires consistent inspection and monitoring of both infrastructure and rolling stock. Traditional inspection practices, while established, remain labour-intensive, time-consuming, and prone to human error, which can undermine operational reliability. Computer vision has become a key technology for automating these processes by enabling accurate visual assessment of tracks, components, and operational environments. Developments in deep learning have significantly strengthened these capabilities by improving defect detection accuracy, enhancing robustness under varied environmental conditions, and supporting real-time operation. More recently, AI-driven inspection frameworks that incorporate multi-sensor fusion, edge computing, and transformer-based architectures have pushed railway monitoring towards predictive, scalable, and increasingly autonomous maintenance solutions. This review provides a comprehensive and structured synthesis of research spanning track inspection, rolling stock monitoring, and passenger and operational safety. It evaluates recent methodological advances, summarises the strengths and limitations of current approaches, and identifies ongoing challenges related to data availability, computational constraints, and deployment in dynamic outdoor settings. The evidence indicates that track inspection technologies are closest to large-scale practical adoption, while rolling stock and safety monitoring systems are developing rapidly but still require further refinement for widespread deployment. By offering an integrated assessment of the current technological landscape and outlining opportunities for future research, this review supports researchers, industry practitioners, and policymakers in progressing towards safer, more efficient, and more sustainable railway systems enabled by modern computer vision and AI techniques.

Keywords Computer vision · Deep learning · AI-driven systems · Autonomous railway inspection · Real-time monitoring · Predictive maintenance · IoT-based monitoring · Fault detection · Railway engineering · Safety monitoring

1 Introduction

Railways remain a fundamental mode of transportation and continue to play a central role in industrialised economies by enabling the large-scale movement of goods, passengers, and resources across extensive distances [1, 2]. Global railway networks support high transport capacity, reduce road congestion, and contribute to long-term economic growth. Recent reports from the European Union Agency for Railways (ERA) and the International Union of Railways (UIC) highlight increasing maintenance demands associated with growing traffic volumes, ageing infrastructure, and a rise in safety-critical incidents. These reports also emphasise that timely identification of track defects, rolling stock faults, and operational safety risks remains a major challenge, with inspection backlogs posing direct risks to network reliability

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and service continuity. Rail transport also offers sustainability benefits, including lower carbon emissions and higher energy efficiency compared with many alternative transport modes [2–4]. As modern railway networks expand and become more complex, the need for scalable, high-frequency inspection and monitoring strategies has grown accordingly [5, 6].

A significant challenge lies in the continuous inspection and maintenance of infrastructure elements such as tracks, sleepers, fasteners, ballast, switches, and signalling systems, as well as rolling stock components including bogies, wheels, brakes, and traction equipment. Traditional inspection procedures for these assets are predominantly manual and labour-intensive. They are often slow, inconsistent, and vulnerable to human error, particularly under environmental conditions such as low visibility, variable lighting, and adverse weather [7]. Maintaining consistent inspection standards across geographically distributed rail networks requires substantial time and resources, which can limit the ability to detect defects early and reliably.

Safety remains a critical concern, since undetected faults such as cracked rails, loose or missing fasteners, worn wheels, or obstructed level crossings can escalate into severe incidents [8]. Studies show that these faults continue to contribute to derailments and operational disruptions across many networks worldwide [9–13]. The growing scale of rail operations therefore necessitates inspection technologies that can provide frequent, accurate, and comprehensive assessments of asset health [14–16].

Computer vision has emerged as a promising solution for addressing these needs by automating key inspection and monitoring tasks. Advances in high-resolution imaging, deep learning, and real-time processing allow computer vision systems to detect and classify a wide range of defects across both track infrastructure and rolling stock with high reliability [17–21]. Convolutional Neural Networks have demonstrated strong capability in identifying structural defects such as cracks and surface wear [22–24]. Object-detection models in the YOLO family support real-time identification of anomalies during train operations [25, 26]. Vision Transformers, which capture long-range spatial dependencies, offer additional advantages for large-scale inspection datasets [18, 27]. Beyond immediate defect detection, computer vision also supports temporal analysis of wear patterns and fatigue indicators, enabling predictive maintenance strategies that reduce downtime and improve reliability [28, 29].

The adoption of computer vision in railway engineering has increased rapidly over the past decade. Publication trends across major research databases show a sustained rise in research outputs since 2015, driven by advances in imaging hardware, improved deep learning architectures, and updated inspection requirements established by operators

and regulators. Several systems have progressed from research prototypes to early-stage deployment, including high-speed bolt inspection platforms such as VISyR, UAV-based corridor surveys, and onboard rolling stock monitoring systems implemented in parts of Europe and Asia. This growing industrial uptake demonstrates the increasing readiness of computer vision technologies and highlights the need for a consolidated assessment of their capabilities, limitations, and practical deployment requirements.

This review synthesises research on the use of computer vision in railway engineering, with particular emphasis on automated inspection, condition monitoring, and maintenance. Early applications primarily focused on rail surface defect detection, whereas more recent developments extend to fastener inspection, wheelset monitoring, sleeper crack detection, valve state assessment, and complex subsystem analysis [6, 7]. The studies summarised in Table 1 illustrate diverse approaches including YOLO-based object detection [20], Vision Transformers [18], and attention-enhanced lightweight networks [19]. The review also examines systems that expand spatial and operational coverage, such as UAV-based surveys [30] and stereo vision approaches for bolt monitoring [31]. These developments collectively support more predictive maintenance strategies and contribute to improved inspection frequency and network reliability.

The objective of this review is to provide a comprehensive assessment of how computer vision supports the development of intelligent and resilient railway systems. The review examines both the strengths and limitations of existing solutions and offers guidance for researchers and practitioners seeking to improve inspection accuracy, operational scalability, and real-world deployment readiness. Although several recent surveys address artificial intelligence in railways [6, 32], they tend to focus on narrow topics such as autonomous train operation or limited maintenance areas. This review provides a unified perspective that spans track inspection, rolling stock monitoring, obstruction detection, and passenger safety. It also highlights the increasing integration of multimodal sensing technologies, including RGB, LiDAR, and thermal imaging, supported by edge computing and transformer-based models. By comparing methodologies across applications and analysing deployment challenges, this review aims to bridge the gap between research developments and practical implementation, supporting continued progress towards intelligent, data-driven railway maintenance and operation.

2 Methodology of the review

To provide a comprehensive and technically grounded overview of computer vision applications in railway engineering, a structured literature review methodology was adopted. The

Table 1 Summary of computer vision methods for railway component inspection

Component	Method	Input data	Performance metrics	References
<i>Train components detection (valves, bolts)</i>				
Train bolt inspection	BoltVision: convolutional neural network (CNN), vision transformer (ViT)	224 × 224 pixel bolt images (12 MP camera)	ViT accuracy: 93%; CNN lower accuracy	[18]
Train valve inspection	FaultNet (multi-phase U-Net segmentation)	High-resolution VTIS images	Dice coefficient: high; fault detection: 97.26%	[33]
Faulty bolts detection	Lightweight CNN with Attention Mechanism	Custom bolt dataset; high-resolution images	Detection accuracy: 96.43%; F1-Score: 96	[19]
Fastener fault detection	YOLO v3 (Darknet-53, FPN)	High-resolution GoPro Hero7 images	mAP: 84.08%; precision: 89%; recall: 95%	[34]
Binocular vision for bolt-loosening	Stereo vision with periodic structure rule	Stereo images of train bolts	Relative error: <1.42%; detection time: <5 min for 8-car train	[31]
<i>Track detection (obstruction, alignment)</i>				
Vegetation and obstacle detection	Faster R-CNN (Inception v2/ResNet v2 Atrous)	UAV video footage of tracks	mAP: 55%–66%; IoU: 0.1–0.4; PR curve used	[30]
Obstacle detection (Stationary/Moving)	Background subtraction, optical flow, Sobel edge detection	RGB video frames (dynamic obstacles)	Sensitivity: high; dynamic obstacle detection qualitative	[35]
Rail track condition monitoring	Edge extraction, morphological processing	Train-mounted RGB camera images	Qualitative segmentation performance	[36]
<i>Fasteners and sleepers detection</i>				
Concrete crack detection	VGG-16, VGG-19, ResNet50, etc	Image patches of concrete cracks (varied datasets)	Accuracy: VGG-16 = 99% (small datasets)	[21]
Rail surface flaws (Corrugations, Squats)	Gabor filtering for texture analysis	High-resolution rail surface images	Texture pattern detection: Qualitative performance	[20]
Railway sleepers defects detection	CNN-based automated detection system	Concrete sleeper crack dataset (open-source)	Detection accuracy: 92%–96% (small cracks identified)	[37]
<i>Safety monitoring</i>				
Passenger safety monitoring	CNN	CCTV video frames from railway stations	Confusion matrix, ROC curve, precision/recall	[38]
Trespassing detection	Mask R-CNN, ROI methods	Live surveillance video streams	Detection accuracy: 100% (field tested)	[39]
Grade crossing near-miss detection	Customised AI-based algorithm	Surveillance video from grade crossings	Accuracy: high; event identification precise	[40]
Rail surface defects detection	YOLOv3, ResNet	High-resolution RGB images (PASCAL VOC dataset)	Detection rate: 97%–100%; time: 0.15 s/image	[41]
Railway behaviour monitoring	YOLO-Pose with ACTION-Net	Crowd behaviour data; 3D simulation	High accuracy in behaviour detection (falls, fights)	[42]
Railway ballast degradation analysis	Machine vision-based degradation analysis	Ballast images from UAVs	Qualitative results in assessing ballast permeability	[43]

aim was to synthesise current research on how computer vision is used to support inspection, monitoring, and maintenance across various railway domains, including track, rolling stock, and station environments. Moreover, this review seeks to:

- Provide a comprehensive synthesis of recent research on the application of computer vision across key components of railway systems, including tracks, fasteners, sleepers, rolling stock, and passenger environments.
- Compare various computer vision models such as CNNs, YOLO, Mask R-CNN, and transformer-based

frameworks in terms of their performance, detection capabilities, and implementation contexts within railway infrastructure.

- Highlight the practical advantages and limitations of computer vision techniques in real-world railway settings, particularly in relation to environmental challenges, dataset constraints, and hardware limitations.
- Identify key opportunities for integrating computer vision with existing remote condition monitoring (RCM) systems and IoT-enabled sensor networks to create more adaptive and intelligent railway operations.

- Present future directions and research gaps, offering insights into promising areas for further innovation and real-world deployment.

2.1 Search strategy and databases used

A structured and comprehensive literature search was conducted to identify relevant studies for inclusion in this review. The search strategy focused on sourcing peer-reviewed publications and preprints that addressed the application of computer vision technologies in railway engineering. The primary goal was to gather a wide spectrum of research that spans both theoretical advancements and real-world implementations in inspection, monitoring, and safety systems.

The search was performed across several major academic databases known for their coverage of engineering, computer science, and transportation research. These databases included:

- *IEEE Xplore* for papers focused on engineering applications and hardware-level solutions.
- *ScienceDirect (Elsevier)* to include multidisciplinary research on computer vision and machine learning, transportation, and railway maintenance.
- *SpringerLink* to capture publications from journals and conferences related to intelligent transportation systems and robotics.
- *MDPI* for open-access research on sensors, automation, and railway technologies.
- *Google scholar* to ensure broad coverage and inclusion of grey literature and highly cited preprints.
- *arXiv* for recent developments and preprints related to deep learning, computer vision, and railway-specific AI applications.

The search utilised Boolean logic and keyword combinations to maximise relevance. Key phrases included: “*railway inspection*”, “*computer vision in railways*”, “*rail defect detection*”, “*rolling stock monitoring*”, “*deep learning for railway safety*”, “*automated railway inspection*”, “*object detection in railway systems*”, and “*railway track monitoring using AI*”. The initial search yielded several hundred papers, which were screened based on relevance, technical depth, and recency. Duplicate studies and non-technical papers were excluded. Priority was given to research articles, conference proceedings, and high-quality review papers that presented empirical results, novel methodologies, or practical implementations.

In addition to identifying relevant works, we analysed their temporal distribution to understand how research attention towards different railway applications has evolved over time. The publication date range was set from 2006 to 2024

to capture a historical perspective while emphasising recent progress. Special focus was placed on studies published between 2018 and 2024 to reflect the growing influence of AI-driven technologies, such as YOLO, Mask R-CNN, and Vision Transformers, in railway inspection and monitoring tasks. This search strategy ensured a balanced representation of foundational research and cutting-edge innovations, forming a solid basis for analysing current capabilities and future directions of computer vision in railway engineering. To further illustrate the evolution of research activity, Fig. 1 presents the annual distribution of publications included in this review across the five application domains. The trend shows a pronounced increase from 2016 onwards, coinciding with the adoption of deep learning architectures and higher industry demand for automated and scalable inspection technologies. Track-related studies dominate the early years, while rolling stock, safety, and obstruction detection have expanded notably in the last five years. This upward trajectory highlights both the accelerating research interest and the need for a consolidated review to summarise existing progress and identify remaining challenges for real-world deployment.

2.2 Inclusion and exclusion criteria

To maintain technical relevance and ensure that the review was grounded in practical applications, a structured set of inclusion and exclusion criteria was applied during the screening and selection process. These criteria helped refine the scope of the review and focus on studies that provide actionable insights into the use of computer vision in railway engineering.

Inclusion criteria:

- *Peer-reviewed journal articles and conference papers* Only scholarly work published in reputable venues was considered, ensuring scientific rigour and peer validation.
- *High-quality preprints* Preprints from platforms like arXiv were included if they presented substantial experimental results, model architectures, or system evaluations related to railway inspection or safety.
- *Focus on computer vision and AI* Studies had to involve the use of computer vision, deep learning, or image processing techniques explicitly applied to railway systems, such as track monitoring, rolling stock analysis, or safety surveillance.
- *Clear application context* Papers that demonstrated a tangible application, either in simulation, field trials, or full-scale deployments, were prioritised.
- *Quantitative performance metrics* Studies were included if they reported accuracy, precision, recall, F1-scores, mAP, or other performance indicators that allowed for objective assessment and comparison.

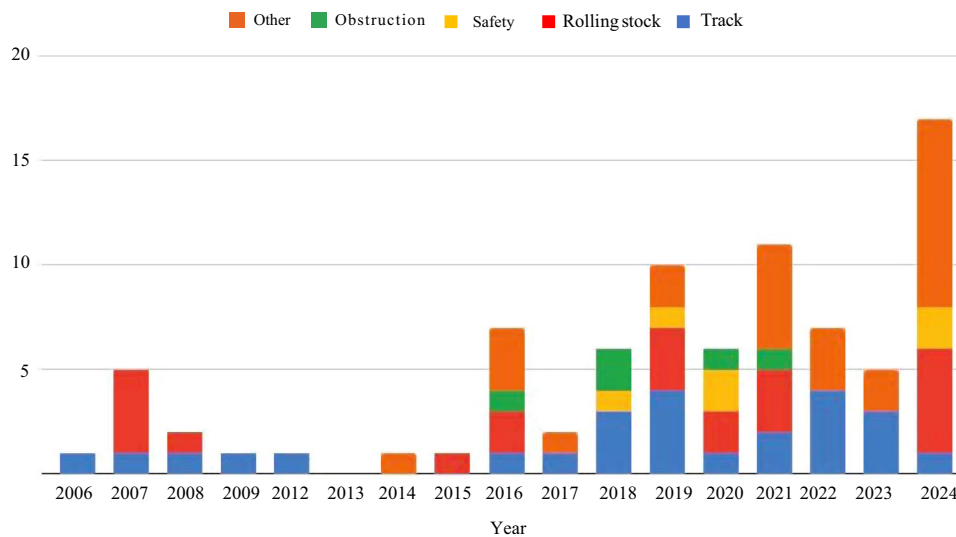


Fig. 1 Annual distribution of publications across major railway computer vision domains (2006–2024). The data show significant growth after 2016, particularly in track, rolling stock, and safety-related studies

Exclusion criteria:

- *Purely theoretical or conceptual works* Papers that proposed frameworks or theoretical models without any implementation, simulation, or experimental validation were excluded.
- *Non-relevant domains* Studies focused on non-railway applications of computer vision, such as road traffic, aviation, or maritime, were excluded unless their methodology was directly transferable to railway systems.
- *Lack of technical depth* Articles with minimal detail on methodology, dataset, or model architecture, or those without reproducible results, were not considered.
- *Duplicate publications* In cases of multiple papers presenting the same dataset or methodology by the same authors, only the most comprehensive or recent version was included.
- *Outdated methods* Studies relying on obsolete techniques (e.g. simple thresholding or edge detection without any learning component) were filtered out unless they served as comparative baselines for more modern approaches.

This filtering process ensured that the final selection of papers reflected both the state-of-the-art in computer vision and its relevance to the evolving challenges within railway engineering.

2.3 Paper classification

To facilitate structured analysis and ensure comprehensive coverage, the selected papers were categorised into thematic domains based on their primary application focus within

railway engineering. This classification allowed the review to compare techniques, challenges, and trends within each specific area of application.

- *Railway track inspection* This category includes studies addressing the condition assessment of track infrastructure, including detection of rail surface wear, fatigue cracks, corrosion, ballast fouling, fastener condition, sleeper condition and track geometric misalignment. Techniques in this category range from traditional image processing to advanced deep learning approaches like Mask R-CNN, YOLO, U-Net, and Deeplabv3+.
- *Rolling stock monitoring* Papers in this group focus on the health monitoring of moving components such as wheels, bogies, axles, suspension systems, and isolation/control valves. These studies explore the use of both vision-based and multi-sensor fusion approaches for detecting mechanical faults, misalignments, and other critical issues in rolling stock components. The category also includes research employing simulation-driven models and Bayesian networks to manage uncertainty and variability in operational conditions.
- *Safety and obstruction detection* This domain comprises research aimed at ensuring passenger and operational safety through the detection of obstacles, trespassers, and abnormal human behaviours near or on railway tracks and stations. It includes small-object detection (e.g. tools, helmets), trespass detection at grade crossings, fall/slip/trip event monitoring in stations, and crowd density estimation. Techniques such as CNNs, DFF-Net, Mask R-CNN, YOLOv5, and transformer-based models have been widely adopted in this category.

- **Challenges and future directions** This category includes studies that analyse the broader challenges of deploying computer vision in real-world railway systems. Topics include computational constraints in edge devices, environmental variability, data diversity, real-time system integration, and the fusion of computer vision with other remote condition monitoring (RCM) sensors, internet of things (IoT), and field-programmable gate arrays (FPGAs). It also covers proposals for next-generation AI architectures and suggestions for future research trajectories.

This thematic classification provides a structured foundation for comparing methodologies, datasets, and deployment readiness across diverse application areas, enabling a clearer understanding of the state-of-the-art and identifying key gaps for future innovation.

2.4 Review scope and summary

This review encompasses a comprehensive analysis of over 70 scholarly articles spanning the period from 2007 to 2024, with an emphasis on recent developments from 2018 onwards. The selected studies were systematically examined for key technical and operational aspects, including model architecture, methodology, input data type (e.g. RGB, thermal, LiDAR), dataset characteristics, performance metrics (e.g. accuracy, precision, recall, mAP, FPS), and deployment feasibility. Special attention was given to works that demonstrated significant potential for practical implementation in operational railway environments. These included models integrated into real-time systems, studies involving in-service or field testing, and methods designed for edge deployment or compatible with existing Remote Condition Monitoring (RCM) infrastructure. The review also prioritised papers that offered insights into predictive maintenance strategies or showcased innovations in safety-critical applications, such as trespassing detection, passenger

flow monitoring, and autonomous inspection platforms. A temporal review of publication trends across the period 2007–2024 was also conducted to understand how research activity evolved within the four major application domains. These trends are described narratively rather than presented graphically, since early studies are sparse and categories frequently overlap, which could lead to misleading visual interpretation.

2.5 Camera systems and image acquisition setups in railway inspection studies

A variety of imaging and sensor systems have been employed across the reviewed studies for railway inspection and safety monitoring tasks. High-speed RGB cameras mounted on inspection vehicles were commonly used for fastener and rail surface monitoring, with resolutions ranging from 2 to 12 MP [19, 34]. UAV-based systems typically employ high-resolution RGB cameras (e.g. GoPro Hero7 or similar) for aerial surveys of tracks and vegetation [30]. Stereo vision systems, such as those used by Sun et al. [31], combine two cameras to capture depth information critical for bolt-loosening detection. Specialised imaging modalities, such as infrared and thermal cameras, were less common but were suggested to enhance defect detection under low-light conditions [44].

Image acquisition platforms varied from train-mounted systems capable of capturing images at speeds up to 201 km/h [45] to drone-based systems operating at moderate altitudes to cover large track sections. High-speed cameras (120 FPS or higher) were often necessary for monitoring bogies and wheels in motion [46]. Additionally, edge computing platforms, such as NVIDIA Jetson Nano and FPGA boards, were frequently integrated to support real-time data acquisition and processing on mobile inspection units. As can be seen in Table 2, these diverse imaging setups highlight the importance of sensor selection and

Table 2 Summary of camera systems and image acquisition platforms in railway inspection studies

Platform type	Camera type	Processing setup	Application area	References
Train-mounted systems	High-speed RGB cameras (2–12 MP)	Mounted on maintenance vehicles; captures at high speeds (up to 201 km/h)	Fastener inspection, rail surface defect detection	[19, 34, 45]
UAV (Drone) systems	High-resolution RGB cameras (e.g. GoPro Hero7)	Drone flights over track sections at moderate altitudes	Vegetation detection, track alignment monitoring	[30]
Stereo vision systems	Paired RGB cameras	Stereo rigs for depth estimation; mobile or static setups	Bolt-loosening detection with depth information	[31]
Thermal/infrared cameras	Specialised thermal imaging sensors	Mounted on trains or UAVs; captures thermal anomalies	Low-light and hidden defect detection	[44]
Edge platforms	Integrated NVIDIA Jetson, FPGA-based setups	Onboard real-time image acquisition and local processing	Real-time inspection of fasteners, track, and rolling stock	[45, 46]

system design in achieving accurate and reliable railway inspections.

To move beyond a linear listing of individual studies and provide a more analytical perspective, it is important to compare the underlying model architectures that dominate railway computer vision research. Although many works report strong performance for specific tasks, their practical suitability varies significantly depending on computational constraints, inspection speed, data availability, and environmental variability. Table 3 therefore presents a comparative analysis of mainstream computer vision architectures commonly adopted in railway engineering, highlighting their relative strengths, limitations, and trade-offs in real-world deployment scenarios. This architecture-centric view provides a unifying framework for interpreting the diverse studies reviewed in subsequent sections and clarifies why certain models are favoured for specific railway inspection tasks.

3 Applications of computer vision in railway track inspection

Computer vision technology is revolutionising railway maintenance by enabling real-time, automated inspections that enhance the reliability, accuracy, and efficiency of track monitoring. Railway fasteners and sleepers, which are essential for maintaining track stability and ensuring safe train operations, are now a critical focus for these advancements. Fasteners secure the rails to the sleepers, while sleepers and ballast provide structural support, absorb dynamic loads, and maintain track alignment. The condition of these components is crucial for preventing derailments and other safety hazards. Through the integration of deep learning and image processing techniques, computer vision models can detect wear, cracks, and structural issues in rail infrastructure with remarkable precision. These systems have expanded their scope to include fastener inspection, identifying missing or damaged components, and monitoring sleeper conditions using RGB and thermal imaging. In addition, they offer solutions for rail alignment assessments and obstruction detection, often employing mobile platforms like drones and onboard cameras for comprehensive data collection. This section explores these advancements, focusing on applications such as rail surface damage, fastener inspection, sleeper monitoring, and track alignment assessments. By automating these inspections, railway systems can reduce costly manual checks, improve operational safety, and achieve scalable, cost-effective maintenance strategies. These technologies hold the potential to significantly enhance the efficiency and safety of modern railway operations.

3.1 Fastener detection

One of the significant advancements in fastener inspection comes from Shang et al. [47], who introduced the fastener inspection and quantisation (FIQ) system based on a mask R-CNN architecture. The FIQ method integrates multiple features, including a fastener convolutional block attention module (FCRM) and channel-wise knowledge distillation (CWD), which collectively enhance the segmentation accuracy of fasteners. With a segmentation accuracy of 96.01% and fault detection accuracy of 95.13%, the FIQ system outperforms traditional methods, achieving precise detection and quantification of fastener conditions, as seen in Fig. 2. However, the model's performance is hindered by the challenge of imbalanced datasets, where defective fasteners are less frequent, and real-time processing remains a key area for improvement. Another approach, proposed by Zhuang et al. [49], utilised a two-stage detection system designed to operate in near real-time for railway component inspection, including bolts and clippers. The method combines a Squeeze and Excitation Participated YOLOv3 (SE-YOLOv3) model with a domain-logic-based hybrid model (DLHM) for calibration. This combination achieves a mean average precision (mAP) of 95.2%, effectively identifying missing fasteners and improving overall safety. Despite its high accuracy, the model's reliance on a specific dataset limits its generalisability, and incomplete object shapes in images reduce detection accuracy for smaller components.

YOLOv3, a popular model for object detection, was utilised by Lin et al. [34] to detect fastener faults. The authors developed a high-speed video system mounted on a maintenance vehicle to capture fastener images and trained the YOLOv3 model to identify defective fasteners. The model achieved a precision of 89% and a recall rate of 95%, meeting practical requirements for railway inspection. An innovative virtual detection circle (VDC) method was also introduced to enhance GPS accuracy in locating defects. However, the system's performance is limited by image quality, especially under low-light conditions, where image clarity affected detection reliability. In another study focusing on railway fastener classification, Ou et al. [51] developed a model that categorises fasteners as normal, partially worn, missing, or covered. Using a combination of latent dirichlet allocation (LDA) for feature extraction, conditional random fields (CRF) for segmentation, and a Bayesian hierarchical model for classification, this system achieved an impressive accuracy rate of 97.81%. However, the model's performance depends heavily on segmentation accuracy, which can be affected by lighting variations and fastener orientations, highlighting the need for further robustness under diverse weather conditions. Marino et al. [45] presented VISyR, a real-time visual inspection system for detecting hexagonal-headed bolts in the fastening system

Table 3 Model architecture comparison for railway computer vision applications

Model/architecture	Strengths in railway context	Limitations/trade-offs	Best-fit railway scenarios	Representative references
YOLO family (one-stage detectors)	High throughput with low latency; strong for frequent component detection and defect screening; suitable for continuous inspection pipelines on moving platforms	Can miss very small, low-contrast, or heavily occluded defects if input resolution is constrained; performance may degrade under domain shift (weather, glare, ballast clutter) without targeted augmentation/adaptation	Real-time fastener/bolt detection, sleeper crack screening, rail surface defect detection, onboard/wayside monitoring under motion constraints	[25, 34, 41]
Faster R-CNN / two-stage detectors	Robust region proposals and localisation in cluttered scenes; often strong precision when background complexity is high	Higher compute and latency than one-stage detectors; less suitable for strict real-time constraints; tuning needed for class imbalance and rare-event hazards	UAV corridor survey (vegetation/obstacles), trackside monitoring where throughput is secondary to localisation stability	[30]
Mask R-CNN / instance segmentation	Pixel-level delineation enables defect quantification (extent/shape) and component geometry extraction; supports measurement-driven inspection	High annotation cost (masks) and heavier inference; sensitive to illumination variability unless trained with diverse data or domain adaptation; deployment cost is higher	Surface flaw segmentation, crack area estimation, precise component boundary extraction for maintenance decision support	[39, 41, 47]
U-Net and segmentation variants (semantic segmentation)	Effective for continuous defect regions; strong for structured surfaces and texture-driven anomalies; supports dense prediction for rail/sleeper/ballast analysis	Requires pixel-level ground truth; may over-segment under noise, shadows, and reflections; compute can be non-trivial for high-resolution imagery	Rail surface crack/flaw segmentation; ballast region analysis; track component segmentation as a preprocessing stage	[33, 48, 49]
Vision Transformers (ViT) / hybrid CNN–Transformer	Captures long-range context and global dependencies; can improve discrimination when local appearance is ambiguous; useful for fine-grained component state classification	Data-hungry and memory intensive; may underperform with small, homogeneous datasets; optimisation/distillation often required for edge deployment	Fine-grained bolt/fastener state classification; multi-scene inspection where context improves robustness across viewpoints/backgrounds	[18, 50]
Classical + probabilistic pipelines (e.g. LDA/CRF/FSM, edge/morphology)	Interpretable and sometimes data-efficient; can be effective with limited labels; predictable failure behaviour and easier validation for safety cases	Less robust to significant appearance variation; brittle under environmental changes; limited scalability to new defect types and complex scenes compared to deep models	Legacy deployments, constrained environments, or as a lightweight pre-filter before deep models; rule-based fastener state classification	[35, 36, 51]
Self-/semi-supervised anomaly detection (AE/VAE/contrastive pretraining)	Reduces reliance on labelled defects by learning representations from unlabelled "normal" data; aligns with rarity of railway faults and high cost of annotation	Calibration of anomaly scores is challenging; can over-flag benign changes (dirt, reflections); requires careful definition and curation of "normal" operating data	Early warning screening for novel/rare defects; pre-filtering large operational datasets before targeted supervised detection	[52, 53]
Generative augmentation (GANs / diffusion-based synthesis)	Can increase coverage of rare fault modes, balance class distributions, and simulate hazardous scenarios that are difficult to capture operationally	Risk of unrealistic artefacts or domain gap; synthetic data can bias models if not validated; governance needed for safety-critical use	Augmenting rare crack/fastener failure cases; stress-testing models under adverse conditions (weather, blur, occlusion)	[54]

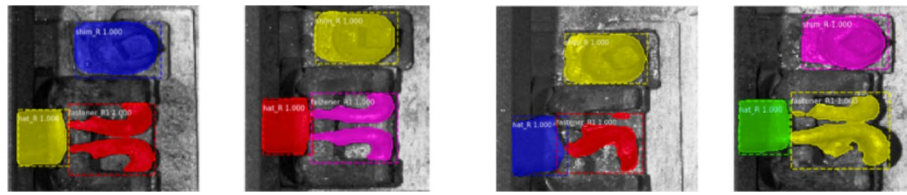


Fig. 2 Detection results of railway fastener components using Mask FRCN, with segmented regions highlighting shims, hats, and fasteners in various colours, demonstrating the model's capability in accurately identifying and localising fastener parts on the track [47]

between rails and sleepers. Utilising a combination of DWT preprocessing and neural network classification on an FPGA platform, VISyR achieved high detection accuracy at speeds up to 201 km/h. While effective, occlusions and environmental variability remain challenges, suggesting the need for further advancements to enhance adaptability.

In summary, these deep learning models demonstrate significant advancements in automating railway fastener detection, each with unique strengths in accuracy, processing speed, and adaptability to various conditions. While models like FIQ and SE-YOLOv3 achieve high accuracy and real-time performance, challenges such as imbalanced datasets, lighting variations, and segmentation accuracy under diverse conditions remain areas for improvement. These studies collectively highlight the potential for computer vision to enhance railway safety by automating fastener inspection, paving the way for more efficient, reliable railway maintenance.

3.2 Sleepers and ballast condition monitoring

Zhang et al. [55] proposed a convolutional neural network (CNN) approach, specifically a three-layer convolutional neural network (TLCNN), to detect absent sleeper support caused by voids or insufficient ballast. The model was trained on simulated vertical displacement data transformed into a time-space matrix, achieving high accuracy in identifying sleeper support deficiencies even in noisy environments. The TLCNN's integration of simulated and real data highlights its robustness in detecting issues that may impact track stability. However, the model's effectiveness varies with speed and axle loads, indicating that further improvements in model depth and training data diversity are necessary for complex track conditions. Similarly, Siahkouhi et al. [56] introduced a self-sensing railway sleeper made from carbon nanotube (CNT)-infused concrete for detecting hanging sleeper defects in ballast tracks. This smart sleeper monitor supports conditions by measuring electrical resistance changes in response to load variations. Experimental results showed that a CNT content of 0.5%, with optimised electrode placement, allowed the sleeper to accurately detect unsupported conditions in disturbed ballast. Finite element modelling (FEM) validated the method's applicability under

varying speeds and loads, though the method's sensitivity is influenced by ballast quality and environmental conditions.

Lesiak et al. [58] developed a computer vision algorithm that uses RGB imaging to detect visible signs of ballast degradation, such as vegetation overgrowth and local muddy areas (LMAs). These issues often indicate poor drainage and potential track stability problems. By analysing the colour intensity and applying thresholding techniques, the algorithm effectively identified ballast degradation features associated with vegetation and mud. While this RGB-based method provides a cost-effective approach for detecting visible degradation, its performance depends on stable imaging conditions, and enhancements in robustness are required to handle variable lighting and environmental factors. Franca and Vassallo [57] proposed a computer vision-based system for classifying railway sleepers and detecting surface defects in real-world environments. Their approach combines Haar transforms, edge detection, and entropy computation to identify sleeper types (wooden or steel) and detect defects in wooden sleepers, achieving 97% accuracy for classification and 93% accuracy for defect detection, as shown in Fig. 3. Testing on high-speed imagery captured from a moving vehicle demonstrated the system's resilience under varied lighting conditions, though accuracy was affected by occlusions and obstructions such as vegetation and debris.

Guerrieri et al. [59] introduced a Digital Image Processing (DIP) technique to measure railway track defects and assess ballast gradation using monocular and stereocular vision. The method achieved a close correlation with traditional measurement techniques, with deviations under 5% in ballast gradation assessments. The stereoscopic approach proved particularly effective for continuous in-situ rail profile monitoring, while monocular imaging was more suitable for controlled laboratory environments. However, the method requires controlled lighting and precise camera alignment for accurate measurements, which limits its adaptability to field conditions. Yella et al. [60] developed a computer vision system for the automated inspection of wooden sleepers, achieving 90% classification accuracy in categorising sleepers as good or bad. The system utilises image analysis techniques such as entropy filtering, morphological operations, and edge detection, with classification performed using multi-layer perceptron (MLP) and support

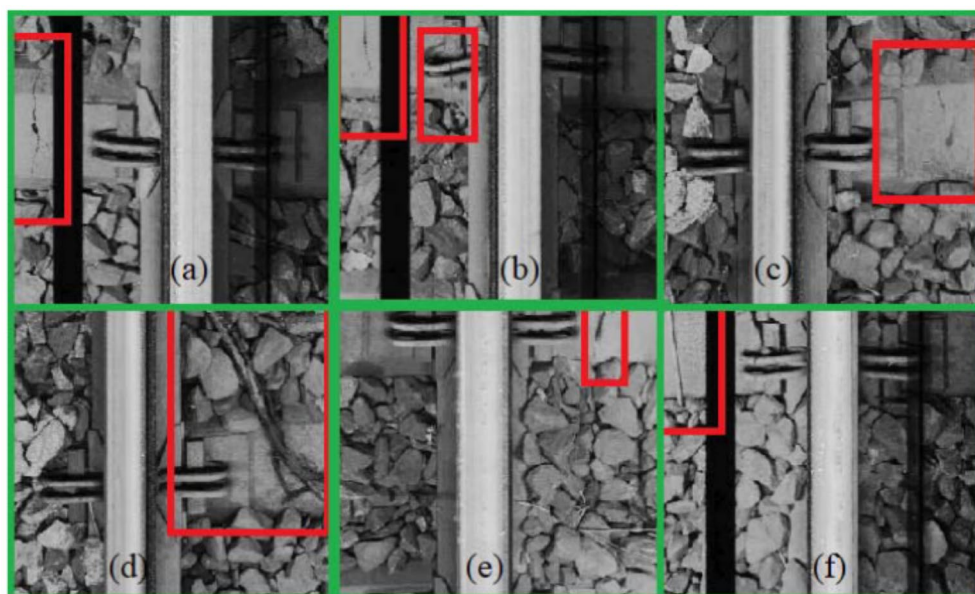


Fig. 3 Examples of various steel sleeper conditions observed in railway monitoring: **a** defective steel sleeper, **b** oil stain on sleeper and fastener, **c** oil slick on sleeper and ballast, **d** vegetation on the sleeper, **e** stone shadow over vegetation on sleeper, and **f** vegetation on sleeper. The image highlights common scenarios encountered in sleeper inspection and condition monitoring [57]

vector machine (SVM) classifiers. The method was tested on 200 sleepers, with feature fusion from both sides of the sleeper enhancing classification performance. While effective, the system's accuracy is affected by environmental conditions, including lighting variability and debris, which limits its reliability under diverse field conditions.

3.3 Rail alignment and geometric inspection

Rail alignment and geometric inspection are critical for maintaining the safety and stability of railway tracks. Misalignment and other geometric defects can compromise track integrity, leading to derailments and other operational hazards. Computer vision and deep learning techniques could become invaluable for monitoring these geometric aspects in real-time and addressing challenges related to environmental factors like lighting, weather, and vibration. This section discusses advanced techniques for detecting geometric defects, tracking alignment changes, and overcoming environmental limitations.

Shi et al. [61] introduced a deep learning-based virtual point tracking system for real-time displacement measurement specifically tailored for railway applications. The proposed method leverages LightPointNet, a lightweight CNN architecture, to monitor the lateral displacement of wheels on rails without the need for physical optical targets. This approach enables high-precision measurement with an accuracy of up to 2 mm, and its real-time capabilities were validated through field tests, achieving over 30 frames per second on edge computing devices. While effective, the system

faces challenges under harsh outdoor conditions, including image noise and motion blur, which limit its generalisability across diverse environments.

Another approach for monitoring rail stability is the interferometric synthetic aperture radar (InSAR) technique, as applied by Zhang et al. [62] to the Qinghai–Tibet Railway. This study utilised the FRAM-SBAS (Full Rank Matrix Small Baseline Subset) time-series method to monitor ground deformation in challenging high-altitude and permafrost-laden environments. InSAR data from 2003 to 2012 revealed significant subsidence, particularly in the Damxung area, with deformation rates reaching 20 mm per year. These insights highlight the technique's ability to detect long-term alignment shifts, although challenges remain in areas with low coherence, such as vegetated regions, and in maintaining data continuity due to temporal gaps.

In addition to satellite-based methods, UAV-assisted inspections have shown promise in geometric monitoring. Banic et al. [63] developed a computer vision-based system for UAV-assisted railway inspection, focusing on track alignment and infrastructure elements. This system employed edge detection and k -nearest neighbours (k -NN) classification to identify and monitor track features such as rails and sleepers from drone imagery. Although effective, the system's performance is sensitive to lighting conditions and depends on high-quality images, which can be difficult to obtain consistently in outdoor environments. Despite these limitations, UAV-based monitoring offers flexible and comprehensive coverage for detecting alignment changes across large track sections.

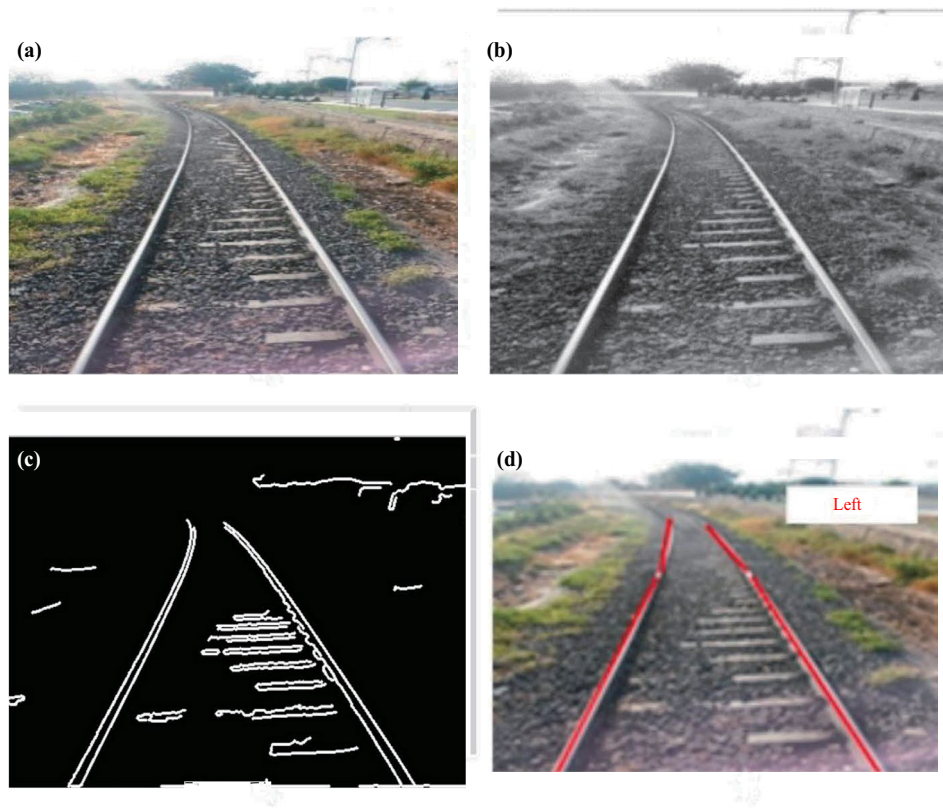


Fig. 4 Illustration of railway direction detection process from Karakose et al. [36]: **a** original railway track image; **b** filtered image; **c** edge-detected image highlighting track features; **d** final direction detection with Hough transform indicating a left turn

Karakose et al. [36] proposed a condition monitoring system that combines edge detection, feature extraction, and decision tree classification to identify geometric defects along the track. As shown in Fig. 4, the system, tested on a vehicle-mounted camera setup, uses multiple cameras to monitor alignment, sleeper conditions, and track direction. By applying image-based measurements and decision rules, this system can track real-time geometric changes at moderate speeds (up to 36 km/h), though high ballast density and poor lighting can reduce accuracy. Recent research has also explored the integration of deep learning-enhanced Kalman filtering techniques for real-time railway monitoring. Wen et al. [64] proposed a residual-based extended kalman filter (REKF), augmented with a deep neural network (DREKF), significantly improving the estimation accuracy of nonlinear train operation states. These developments offer promising directions for enhancing railway track alignment monitoring under dynamic and complex operational conditions.

In summary, advancements in computer vision, radar technology, and AI-enhanced filtering can greatly improve the ability to monitor and maintain rail alignment and geometric stability. Techniques such as virtual point tracking, InSAR, UAV-assisted inspections, and condition monitoring systems provide comprehensive tools for detecting and

addressing alignment issues. However, challenges such as environmental sensitivity, data continuity, and image quality remain. Future research and technological improvements in robustness, generalisation, and real-time system integration are essential to maximise the impact of these inspection methods on railway safety and operational efficiency.

3.4 Autonomous track inspections

Autonomous track inspections leverage deep learning models to detect rail surface wear, fatigue cracks, corrosion, and other structural issues. While traditional methods such as ultrasonics and eddy-current testing remain widely used for crack and surface defect inspection, they require specialised equipment and are typically limited to operating speeds of around 80 km/h. In contrast, deep learning models can identify smaller defects and can be deployed on in-service trains running at full speed, provided the image frame rate is sufficiently high. This capability enables faster and more accessible inspections without additional equipment, improving both the efficiency and scalability of track monitoring.

Hua et al. [17] introduced a robotic-assisted deep learning approach for non-destructive fault detection in bolted rail joints, which are critical structural components of the

track. Their system achieved an average accuracy of 84.8% at 20 FPS by using K-means clustering to optimise bounding box sizes and an adaptive loss function to enhance detection performance. By combining deep learning with a robotic inspection platform, the system accurately identified various bolt faults, including missing or damaged fasteners, and demonstrated robust performance in real-world conditions. Kou et al. [21] presented a deep learning-based rail crack detection model using Deeplabv3+ with a ResNet50 backbone. With an accuracy of 96.4%, the model reliably detected rail cracks under challenging lighting conditions, aided by a region-growth algorithm that reconnects fragmented crack segments. This low-cost method offers a practical alternative to magnetic particle inspection and shows strong potential for routine inspection of rail surfaces across diverse environments.

Another promising model by Guo et al. [41] applied mask R-CNN to rail surface defect detection, achieving mean average precision scores of 65.43% for bounding boxes and 64.50% for segmentation masks. The model outperformed traditional thresholding methods by handling variations in defect severity and rail orientation under different lighting conditions, as illustrated in Fig. 5. Although mask R-CNN

faces challenges with very small defects, it remains a robust and scalable alternative to manual inspection. Xia et al. [37] proposed CF-NET, a high-speed one-stage crack detection model based on RetinaNet. CF-NET achieved 98.1% accuracy at 17 FPS and used a divide-and-conquer labelling strategy to address scale variation and label mismatch issues. While effective for concrete sleeper inspection, some pre-processing demands and empirically defined module parameters limit its cross-domain scalability.

TrackNet, introduced by James et al. [48], demonstrated the ability to reduce false alarms in vision-based track inspection systems by combining U-Net segmentation with DenseNet or ResNet classification. This two-stage approach achieved 90% accuracy, improving the reliability of rail defect detection in comparison with conventional VTIS pipelines. Zhuang et al. [20] extended this concept through a deep learning framework capable of identifying multiple flaw categories, including corrugations and squats, with an accuracy of 98.2%. Their model integrates a rail extractor with cascading flaw classifiers, illustrating how traditional feature extraction and deep learning can be effectively combined for comprehensive rail defect inspection.

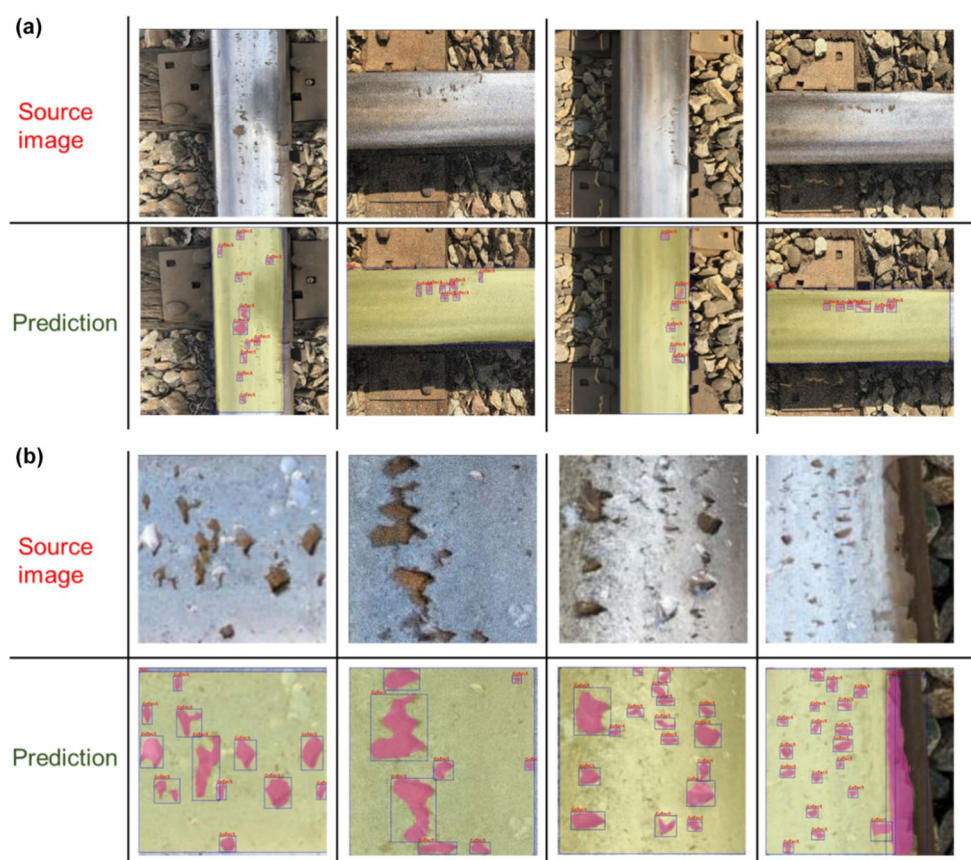


Fig. 5 Inspection results of rail surface defects using Mask R-CNN, showcasing detection under varying severities: **a** mild defect conditions, where small, isolated flaws are identified; **b** severe defect conditions, with extensive flaw coverage highlighted, illustrating the model's segmentation accuracy across defect intensities [41]

Advances in mobile inspection platforms have further accelerated the feasibility of autonomous track monitoring. Karakose et al. [35] developed a train-mounted image acquisition system for real-time rail inspection, employing edge detection and morphological processing to detect pitch, expansion, and surface faults. Although the system's accuracy tends to decrease under severe lighting and environmental conditions, it highlights the significant value of pairing deep learning models with mobile imaging platforms to achieve continuous, real-time track monitoring.

In summary, recent advancements in computer vision and mobile platforms have revolutionised technology developed for the inspection and maintenance of railway tracks, making them more efficient, accurate, and scalable. Techniques utilising models such as Deeplabv3+, Mask R-CNN, and U-Net have showcased remarkable capabilities in detecting surface wear, structural defects, and alignment issues. These automated solutions not only reduce reliance on traditional manual inspections but also enhance safety, operational reliability, and predictive maintenance capabilities across the railway industry. Table 4 provides a comprehensive summary of the applications discussed in this section, highlighting the transformative potential of computer vision in railway track inspection.

4 Obstruction detection and safety monitoring

Ensuring the safety and operational efficiency of railway systems requires addressing two critical aspects: detecting physical obstructions on the tracks and monitoring safety conditions for passengers and railway infrastructure. Obstructions, ranging from small tools to large debris, pose significant risks to train operations, demanding robust and real-time detection mechanisms. Similarly, passenger safety in stations and near tracks is paramount, requiring advanced technologies to monitor crowd behaviour, detect unauthorised access, and predict potential risks. Recent advancements in computer vision and machine learning have enabled the development of sophisticated systems that enhance obstruction detection and safety monitoring capabilities. By integrating high-accuracy models, real-time processing, and multi-sensor data fusion, these technologies offer innovative solutions to improve railway safety. This section explores key advancements in obstruction detection and passenger safety monitoring, highlighting the impact of computer vision in addressing these challenges.

4.1 Obstruction detection

Obstruction detection on railway tracks is crucial for preventing accidents and ensuring the safe and efficient operation of trains. Various computer vision and deep learning techniques have been developed to identify obstacles on the

tracks, ranging from small objects like tools and vegetation to larger obstructions like fallen branches or debris. The following section explores recent advancements in obstruction detection, focusing on object-detection algorithms, environmental challenges, and the integration of real-time monitoring systems for enhanced railway safety.

One approach to improving obstruction detection is the differential feature fusion convolution neural network (DFF-Net), introduced by Ye et al. [66]. Designed to detect small railway objects such as helmets, tools, and other potential hazards, DFF-Net combines a prior detection module with an object-detection module that uses differential feature fusion. This architecture allows DFF-Net to achieve a high mean average precision (mAP) of 90.12% and a detection speed of 54 frames per second, outperforming the SSD model in both speed and accuracy. Particularly effective for small-object detection, DFF-Net demonstrated improvements of 4.61% and 3.29% mAP for helmets and spanners, respectively, compared to SSD. However, the model faces challenges with low-light conditions and is complex for deployment on embedded systems, suggesting a need for further optimisation. Kafetzis et al. [30] explored a UAV-assisted aerial survey system that employs deep learning to detect obstacles and vegetation on railway tracks. By utilising UAVs equipped with faster R-CNN models (Inception v2 and Inception ResNet v2 Atrous), the system achieved satisfactory detection accuracy for various obstruction types, especially vegetation and obstacles close to or on the tracks. Detection precision ranged from 64 to 79% for obstacles and up to 93% for certain types of vegetation, with the UAV system capable of recognising railways with 99% accuracy. Despite its high precision, the model's generalisability across different datasets remains limited, and the processing speed of one frame per second on the NVIDIA Jetson Nano constrains its real-time application potential.

Yu et al. [67] proposed a neural network-based obstacle detection algorithm specifically designed for real-time railway safety applications. By training the network on a diverse set of railway images containing various obstacles, the authors developed an algorithm capable of improving detection speed and robustness in complex environments. Experimental results showed that the model outperformed traditional methods in terms of accuracy and reliability, even under challenging conditions. However, its performance is affected by extreme environmental factors, such as intense lighting or image noise, and could benefit from an expanded dataset for better generalisation across varied conditions. In addition to direct obstacle detection, obstruction identification can also indirectly benefit from techniques designed for other railway maintenance tasks. For instance, computer vision-based ballast analysis, as demonstrated by Huang et al. [43], can provide indirect insights into track conditions that may affect obstruction detection. By quantifying ballast degradation and correlating it with permeability properties,

Table 4 Summary of applications of computer vision in railway track inspection

Model	Dataset used	Detection task	Performance	References
Mask R-CNN (referred to as mask FRCN in their work)	Custom dataset (SFC-type fasteners)	Fastener inspection and fault detection	Segmentation accuracy: 96.01%; Fault detection accuracy: 95.13%	[47]
SE-YOLOv3 + domain-logic-based hybrid model (DLHM)	Custom dataset (1,294 railway images)	Missing fasteners and component detection	mAP: 95.2%	[20]
YOLOv3	Custom dataset (16,677 images from high-speed video)	Fastener fault detection	Precision: 89%; Recall: 95%	[34]
VISyR System with FPGA-based dual neural classifiers	Images of hexagonal-headed bolts	Detection of presence/absence of bolts	99.6% accuracy for visible bolts, 95% for missing bolts	[45]
Latent Dirichlet allocation (LDA) + CRF + Bayesian FSM	Custom dataset (Database1: 400 hand-labelled; Database2: 10,076)	Fastener classification (normal, worn, missing, covered)	Accuracy: 97.81%	[51]
Three-layer convolutional neural network (TLCNN)	Simulated and real track displacement data (60,000 samples)	Detection of absent sleeper support	High accuracy in noisy environments	[55]
Self-sensing concrete sleeper with CNT infusion	Experimental data with varied CNT concentrations	Detection of hanging sleeper defects	Optimised at 0.5% CNT concentration	[56]
RGB-based image extraction algorithm	Computer vision data from the track geometry car (RGB images)	Ballast degradation detection (vegetation, mud)	High accuracy in ballast condition classification	[43]
Image processing with haar transforms and edge detection	High-speed imagery from a moving vehicle	Sleeper classification and defect detection	Sleeper accuracy: 97%; Defect detection: 93%	[60]
DIP technique with monocular and stereocular vision	In-situ ballast and lab rail samples	Rail wear, ballast gradation, track corrugation	Deviation under 5% vs traditional methods	[10]
Machine vision with MLP and SVM classifiers	Images of 200 wooden sleepers (test vehicle)	Condition classification (good or bad)	Accuracy: 90%	[65]
Deeplabv3+ with ResNet50	Custom dataset (380 images)	Crack detection	Accuracy: 96.4%	[21]
DL-enabled CV with Robotics Assistance	Custom Railway Bolt Dataset (4,257 images)	Detection of missing/damaged bolts	84.8% accuracy at 20 FPS	[17]
Mask R-CNN with ResNet101	Custom dataset (1,040 images including augmentation)	Rail surface defect detection	mAP: 65.43% (bbox), 64.50% (masks)	[41]
TrackNet (U-Net + DenseNet/ResNet)	Custom dataset (not specified)	Defect classification, false alarm reduction	Accuracy: 90%	[48]
Rail Extractor + Cascading Flaw Identifier (DenseNet)	Industrial and online sources	Detection of flaw types (corrugations, squats, shelling)	Identification accuracy: 98.2%	[20]
Canny Edge Detection + Morphological Processing	Train-mounted camera images	Fault detection (pitch, expansion, dressage)	Reliable in real-time	[35]
CF-NET (RetinaNet-based one-stage detector)	Custom dataset (54,095 images)	Crack detection on concrete sleepers	Accuracy: 98.1% at 17 FPS	[37]
LightPointNet (CNN) with rule engine	Field test videos (Italy, Lithuania)	Real-time lateral wheel displacement	Up to 2 mm accuracy, 30 FPS	[61]
InSAR (FRAM-SBAS method)	Long-term satellite data (2003–2012)	Ground deformation monitoring	Millimetre-level precision	[62]
UAV-based machine vision System	Drone-captured images	Monitoring track alignment	High accuracy under stable conditions	[63]
Multi-Camera condition monitoring system	Vehicle-mounted real-time images	Alignment and sleeper condition monitoring	Reliable at up to 36 km/h	[36]

Huang's machine vision-based imaging system (MV-BIS) offers predictive data for track maintenance that could help reduce obstructions caused by ballast fouling. Though not a direct obstacle detection method, MV-BIS contributes to railway safety by informing preventive maintenance strategies, which in turn reduce the likelihood of track obstructions.

In summary, the development of obstruction detection techniques has seen significant progress through models like DFF-Net and UAV-based systems, which leverage deep learning to enhance accuracy and processing speed. While each approach has limitations, such as low-light performance issues in DFF-Net, limited generalisability in UAV-based models, and

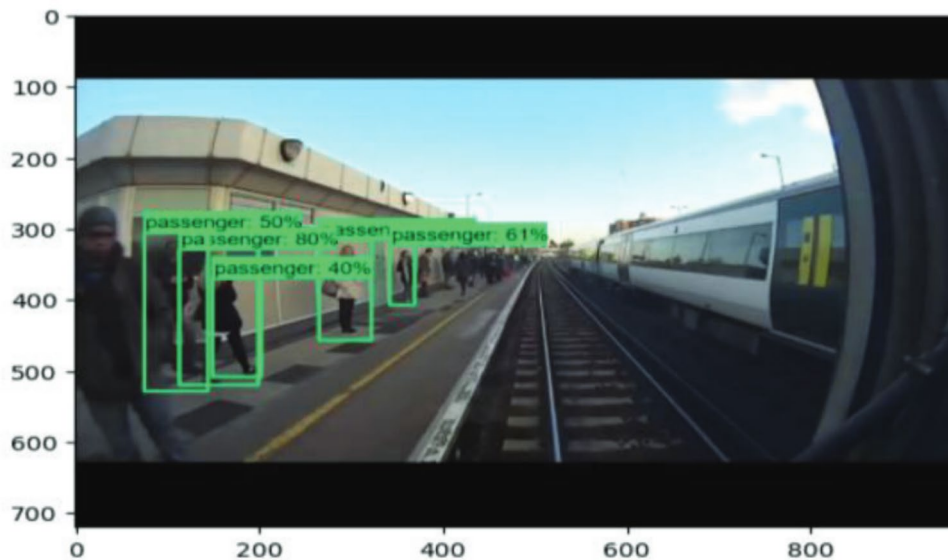


Fig. 6 Passenger detection at a railway station platform using YOLOv5 model, as demonstrated by Gorane and Joshi [68]. The image shows bounding boxes identifying passengers waiting on the platform, each annotated with detection confidence percentages

environmental sensitivity in neural network algorithms, the combination of these methods provides a robust foundation for enhancing railway safety. Recent advancements have also explored sensor fusion techniques, such as combining RGB cameras and LiDAR data for improved railway object detection under challenging conditions [44]. These newer approaches aim to enhance detection reliability and robustness across varied railway environments. Continued advancements in real-time monitoring, embedded system optimisation, and sensor fusion are expected to further improve the practical deployment of obstruction detection systems in railway operations.

4.2 Safety monitoring

Incorporating advanced computer vision technologies, passenger safety in railway environments has become a central focus in recent research. Zhang et al. [42] introduced an intelligent monitoring system for managing large passenger flows in transit stations, employing deep learning and digital twin modelling to provide real-time crowd analysis and abnormal behaviour detection. Their system successfully identifies overcrowded areas and flags high-risk behaviours, such as falling or unauthorised crossing, enhancing station safety management. Similarly, Gorane and Joshi [68] developed a machine learning framework for detecting passengers on platforms and near tracks, improving the automatic warning systems by integrating detection models like YOLOv5 for fast and accurate passenger identification in real-time, as shown in Fig. 6. This real-time object-detection system enhances passenger safety by monitoring platform occupancy and proximity

to the track, enabling automated warnings and preventive safety measures.

Alawad et al. [69] explored the predictive capabilities of machine learning using historical accident data from UK railway stations to identify risk factors for passenger safety, highlighting the efficacy of decision trees for identifying peak accident times and vulnerable passenger demographics. In another approach, Alawad et al. [38] proposed a deep learning framework focused on detecting fall, slip, and trip (FST) events in crowded railway stations using CNNs on CCTV footage, achieving high accuracy in real-time risk detection. Zaman et al. [39] developed an AI-based trespassing detection system using mask R-CNN, achieving 100% accuracy in identifying unauthorised entries across different types of trespassers, such as pedestrians and vehicles, at grade crossings. This approach provides real-time alerts and valuable data for safety measures. Additionally, Zhang et al. [40] focused on detecting near-miss events at grade crossings, which are critical for preventing potential accidents. Their computer vision algorithm, applied to surveillance video, identifies violations of safety protocols in dynamic conditions, such as varying lighting and weather. This system provides data-driven insights for managing safety at high-risk areas like grade crossings.

In summary, computer vision applications for railway safety extend beyond track inspection to encompass comprehensive passenger safety measures. Integrating advanced algorithms for real-time monitoring, behaviour detection, and risk assessment, these systems provide proactive safety solutions in dynamic railway environments. Table 5 summarises the various models, datasets, and performance metrics across these applications, highlighting advancements

Table 5 Summary of computer vision models for obstruction detection and safety monitoring

Model	Dataset used	Detection task	Performance	References
DFF-Net (differential feature fusion CNN)	Custom railway traffic dataset (train-mounted camera)	Small-object detection (e.g. helmets, tools)	mAP of 90.12%, 54 FPS	[66]
Faster R-CNN (Inception v2/ ResNet v2 Atrous)	Two custom datasets (YouTube videos and recordings in Greece)	Vegetation and obstacle detection	Detection precision: 64%–79%, up to 93% for vegetation	[30]
Neural Network-Based Obstacle Detection Algorithm	Custom dataset with railway obstacles	Obstacle detection on tracks	Improved detection speed and accuracy	[67]
Machine Vision-Based Imaging System (MV-BIS)	25 ballast samples (field and lab data)	Ballast degradation and permeability analysis	Strong correlation between PDS and FI	[43]
NF-Net + YOLO-Pose + ACTION-Net	Custom dataset (railway station scenarios)	Crowd density monitoring and behaviour detection	Real-time performance, high accuracy	[42]
EfficientDet and YOLOv5	Public and internet-sourced images	Passenger detection on platforms and near tracks	YOLOv5 F1-score: 38.748, 26 ms detection time	[68]
Decision tree (DT) algorithm	Accident records from UK railway stations	Prediction of accident risk factors	88.7% accuracy, AUC: 0.90	[38]
Mask R-CNN with ROI methods	COCO dataset (training) + surveillance video	Trespassing detection	100% accuracy	[39]
Customised AI algorithm	New Jersey grade crossing video data	Near-miss detection for trespassing	High accuracy; supports proactive safety measures	[40]

in safety monitoring, obstacle detection, and passenger protection in railway stations and grade crossings. These developments in computer vision technology contribute significantly to a safer railway infrastructure, ensuring a more secure experience for passengers and facilitating preventive safety management for railway operators.

5 Vision-based inspection of rolling stock

Efficient and accurate inspection of rolling stock components is essential for maintaining railway safety and ensuring reliable operation. Traditional manual inspection procedures are labour-intensive, time-consuming and prone to human error, which can result in undetected defects in critical components. Recent advances in computer vision and machine learning have enabled automated, high-precision inspection of rolling stock elements, including wheels, bogies and suspension systems, as well as smaller components such as valves and bolts. This section reviews these technological developments and highlights recent research that employs high-speed imaging, deep learning models and advanced algorithms for fault detection and component tracking in railway environments.

5.1 Wheel and bogie fault detection

Shaikh et al. [46] introduced a deep learning-based system that applies the anchor-free YOLOv8 (AFYv8) model for real-time identification and tracking of bogie components, including axles, suspension elements and connecting rods. Combined with high-speed action cameras, the AFYv8 model achieved

a mean Average Precision of 99.1% and demonstrated a substantial improvement in recognition accuracy compared with earlier classifiers. As illustrated in Fig. 7, the system performed reliably across a range of lighting conditions and is capable of detecting missing or deformed parts when bogies move at speeds between 10 and 40 km/h. Remaining challenges include object overlap and reduced precision during inspection of densely packed components. Wu et al. [70] developed a Bayesian deep learning framework for identifying unexpected faults in high-speed train bogies. The model incorporates Bayesian uncertainty estimation to distinguish between known and unknown faults, allowing it to detect complex or concurrent fault patterns. The approach achieved improved mean Average Precision values and demonstrated strong capability for handling out of distribution samples. Despite these advantages, its computational requirements may limit real-time deployment without hardware acceleration. Kou et al. [71] proposed a multidimensional end-to-end convolutional neural network for diagnosing faults in rotating devices on high-speed train bogies. The model processes multidimensional vibration signals directly, removing the need for manual feature engineering, and achieved an accuracy exceeding 98% across laboratory and real-world datasets. Its ability to analyse large volumes of multichannel sensor data makes it suitable for real-time applications in components such as wheels and axles, although field deployment may require dedicated hardware to manage the computational load associated with multidimensional tensor processing.

Chen et al. [72] employed a capsule network (CapsNet) for diagnosing single and compound faults in high-speed train bogies. The model achieved an accuracy of 96.65% and



Fig. 7 Detection of bogie components including axle box, primary suspension spring, binding screw, support beam and support rod using an anchor-free YOLOv8 (AFYv8) model, adapted from Shaikh et al. [46]

outperformed conventional convolutional neural networks in distinguishing compound fault patterns. The vector-based capsule representation enabled the extraction of spatial hierarchies within the data, improving fault discrimination. However, the computational demands of CapsNet present challenges for real-time deployment unless supported by optimised hardware. In addition, the use of simulated data for training may limit generalisability to real-world operating conditions. Dumitriu [73] proposed a signal-based approach for detecting defects in railway vehicle suspension dampers by analysing cross-correlation coefficients derived from vertical bogie accelerations. The method reliably separated faulty and functional dampers in both simulated and field test environments and provided a non-intrusive and cost-effective diagnostic tool. Its performance, however, may be affected by variations in track quality or the presence of significant environmental noise. Prasad and Kishore [74] evaluated the performance of active contour models for segmenting rolling stock components from high-speed video data, using techniques such as the Chan Vese model with shape invariance (CV SI). Their approach achieved a structural similarity index of 0.953, demonstrating strong segmentation accuracy under variable lighting conditions. The use of high-speed cameras and contrast enhanced frames aided segmentation performance, although the iterative nature of active contour models introduces computational overhead that may limit feasibility for real-time applications.

Similarly, Kishore and Prasad [75] developed an intelligent monitoring framework for railway rolling stock using a shape prior-based active contour model (SP CV AC). This method tracks and segments critical bogie components, such as binding screws, and achieved an average similarity index of 89% when compared with manually segmented ground truth images. By automating the segmentation of moving components, the approach reduces the reliance on manual inspections. However, its dependence on predefined shape priors limits adaptability when encountering newly designed or structurally modified parts. In a related study, Kishore

and Prasad [76] introduced a computer vision system for the inspection of rolling stock components including wheels and suspension elements. Using high-speed camera footage and a shape prior level set model, the system accurately segmented and identified defective bogie components while compensating for motion induced deformation and variable illumination. Despite these strengths, challenges persist when dealing with small or heavily occluded components, as well as scenes captured under low lighting, indicating that further refinement is needed to improve robustness. Singh and Kumar [77] examined human error probability in the maintenance of rolling stock components such as brake discs and wheelsets using the human error assessment and reduction technique (HEART). Their analysis identified several error producing conditions, including time pressure, task complexity, and inconsistent working environments, which increase the likelihood of maintenance related faults. These findings highlight the importance of reducing human involvement in routine inspection tasks by adopting automated vision-based systems for critical rolling stock components.

In summary, vision-based inspection systems have proven highly effective in automating the detection of faults in rolling stock components. From deep learning models like AFYv8 and CapsNet to signal-based methods and active contour models, these technologies enable precise, real-time monitoring under diverse operational conditions. While challenges remain, particularly with real-time processing and environmental variability, these advancements offer significant potential for enhancing railway safety and operational efficiency. The accompanying table provides a concise summary of each model, its detection task, performance, and key advantages and limitations.

5.2 Detection of faulty valves, bolts, and other rail engine components

Inspection and fault detection of rail engine components, including bolts, valves, and various fastening mechanisms,

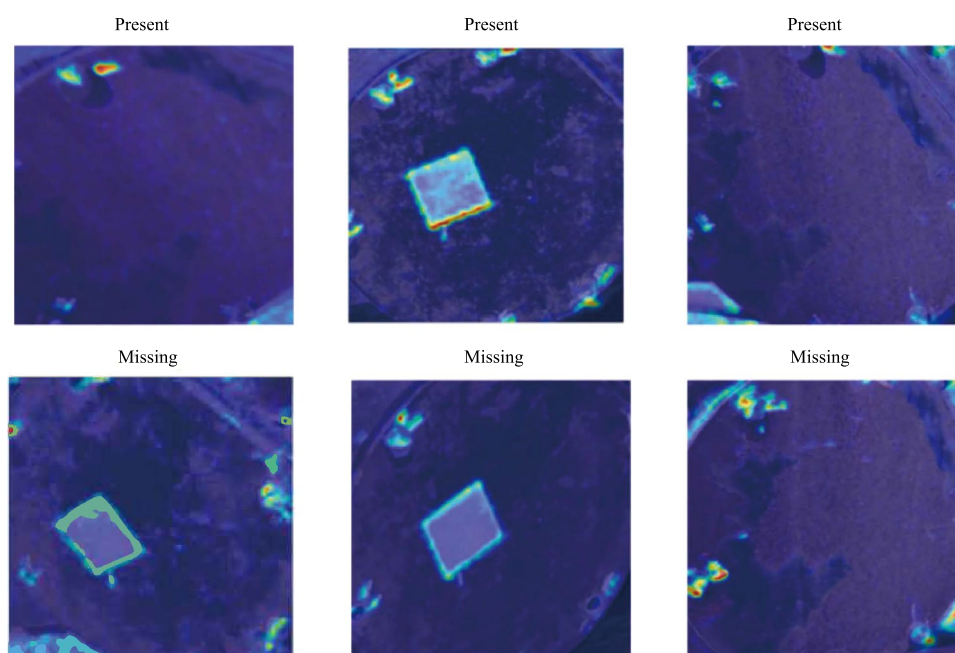


Fig. 8 Visualisation of model predictions using Grad-CAM heatmaps for missing bolt detection by Alif et al. [19]. The top row shows examples where bolts are present, and the bottom row shows cases where bolts are missing

are crucial for ensuring the safety and reliability of railway operations. Manual inspection methods are often time-consuming and inconsistent, and may overlook defects in hard-to-reach or occluded areas. With the integration of robotic systems, computer vision, and deep learning models, automated inspection systems now offer real-time, non-destructive, and precise detection of faults in these essential components. This section reviews several innovative approaches for detecting faulty bolts, valves, and other rail engine components.

In their study, Alif et al. [19] developed a lightweight convolutional neural network with an integrated attention mechanism for real-time missing bolt detection and achieved an accuracy of 96.43%. The attention module effectively focused on the most relevant regions of each image, as illustrated in Fig. 8, where Grad-CAM visualisations highlight the areas that influenced the model's predictions. This interpretability supports greater reliability in practical monitoring tasks. The model also maintained a low computational footprint, making it suitable for deployment on edge devices, although its generalisability could be improved by training on more diverse railway environments. In a related comparative study, Alif et al. [18] introduced the BoltVision framework and evaluated three architectures for missing bolt classification: a convolutional neural network, a compact convolutional transformer, and a vision transformer. The Vision Transformer achieved the highest accuracy at 93% and showed strong potential for deployment in inspection systems. However, its computational complexity presents challenges for low power hardware,

which suggests the need for optimisation techniques such as pruning or quantisation.

Zhou and Huo [78] designed a YOLOv3-based model for detecting delayed bolt fractures in long span steel bridges. Their approach reached an average precision of 89.14% after applying data augmentation to compensate for the limited size of the training dataset. Although effective, the method remains sensitive to lighting variation and bounding box localisation accuracy. Sun et al. [31] proposed a binocular stereo vision system for detecting bolt-loosening in critical components on running trains. Their system achieved a relative error below 1.42% and processed an eight-car train in under five minutes, demonstrating practical feasibility for real-time inspection. Performance may still be affected by extreme lighting conditions, indicating the need for improved robustness in field deployment.

5.2.1 Detection of faulty valves

Pahwa et al. [33] introduced FaultNet, a deep learning-based system designed for the inspection of isolation valves on pneumatic inter-car hose connections used in Singapore MTR trains. FaultNet employs a two-stage U-Net-based segmentation framework and achieved an accuracy of 97.26%, reducing both false positives and false negatives when compared with previous approaches. As shown in Fig. 9, the Grad-CAM visualisations highlight the regions that contribute most to the model's predictions, demonstrating the system's ability to localise and prioritise the

critical areas of each valve. This interpretability supports greater confidence in practical deployment. The authors noted that the model's accuracy may decrease when valves appear at unusual orientations, indicating the need for further improvements in handling geometric variation. Junfeng et al. [79] proposed a vision-based monitoring system for subway ventilation valves that integrates convolutional neural networks with an encoder enhanced virtual coding mechanism. Their method achieved a test accuracy of 98.5% and an F1-score of 99.6%, showing strong capability for real-time valve state recognition while maintaining relatively low computational requirements. The system performed effectively within the tested environment, although additional adaptation is needed to generalise across different valve designs and operational datasets.

Zhou et al. [80] developed a Nanodet-ResNet101-based model for detecting defects in load weigh valves, also referred to as train height valves in their study. The proposed system reached a detection accuracy of 97% and demonstrated real-time processing suitable for onboard inspection. The model successfully identified faults such as missing or misaligned bolts and nuts and outperformed traditional inspection methods. Its performance, however, may vary under extreme lighting conditions, which suggests that further optimisation is required for consistent field deployment.

5.2.2 Inspection of other mechanical components

Earlier studies, such as Liu et al. [65], introduced vision-based systems for inspecting small mechanical components in freight wagons, focusing on axle box lateral retaining keys (BBKs) using SHOG features with SVM classifiers. While effective at the time, these methods struggled with lighting variability and complex freight environments.

More recent research has shifted towards deep learning-based solutions, offering improved robustness and real-time performance. Hua et al. [17] developed a robotic-assisted deep learning system for railway bolt fault detection, achieving an average accuracy of 84.8% in real-world inspections. Similarly, Sun et al. [31] demonstrated a binocular vision-based system capable of detecting bolt-loosening on running trains, with relative errors below 1.42%, enabling real-time deployment. These advancements highlight the trend towards using CNNs, multimodal imaging, and more sophisticated real-time estimation techniques such as deep Kalman filtering, as recently proposed by Wen et al. [64] for nonlinear train operation state estimation, to overcome traditional challenges such as environmental noise, occlusions, and variable lighting conditions.

In addition, thermal management is becoming increasingly important for mobile inspection units and maintenance vehicles operating in extreme railway environments. Intelligent systems like the ITMS proposed by Bhorge et al. [81] offer innovative solutions for efficient thermal control, which could enhance the performance and longevity of railway inspection robots and autonomous monitoring devices. In an earlier study, Liu et al. [82] developed a computer vision system for detecting missing fastening bolts on freight wagon brake systems using SVM classifiers. Although successful at the time, modern approaches now favour deep learning-based methods for higher accuracy and scalability across diverse railway environments.

Overall, recent developments demonstrate significant progress towards scalable, robust, and real-time automated inspection systems for small mechanical components, which are critical for enhancing railway safety and maintenance efficiency. A summary of the papers discussed in this section is provided in Table 6.

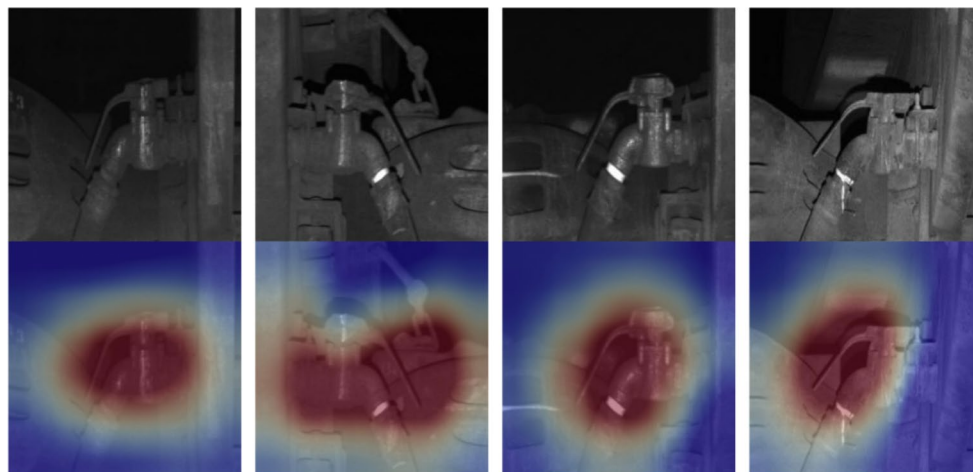


Fig. 9 Heatmaps generated by Grad-CAM for valve fault classification in train engines, as proposed by Pahwa et al. [33]. The top row shows original grayscale input images of valves, while the bottom row visualises the model's focus areas during classification

6 Challenges and limitations

As computer vision technology advances, it has become an integral part of railway inspection and safety systems. However, deploying these systems in real-world railway environments presents numerous challenges that impact their effectiveness and reliability. This section addresses some key challenges related to environmental conditions, real-time processing, computational limitations, and data availability.

6.1 Environmental and lighting conditions

Environmental factors such as weather, lighting, and occlusions significantly impact the accuracy and robustness of computer vision systems in railway applications. Many models, including those by Zhang et al. [68] and Gorane and Joshi [42], require high-quality image inputs for accurate passenger or obstacle detection. Poor lighting in nighttime or overcast conditions, as well as shadows or obstructions from infrastructure or vegetation, can lead to reduced detection performance. Adverse weather conditions such as rain, fog, and snow further degrade image quality, making it difficult for computer vision algorithms to discern fine details.

Studies such as Alawad et al. [38, 69] highlighted the limitations of static CCTV footage in detecting fall, slip, and trip events in dynamic station environments, particularly during rush hours or emergency situations with high crowd density. Similarly, Zhang et al. [40] demonstrated that surveillance-based near-miss detection at grade crossings could suffer from environmental variability, with performance heavily dependent on video quality, lighting, and background noise. To address these challenges, adaptive algorithms are emerging as a promising solution. Examples from other domains, such as autonomous driving and surveillance, demonstrate the feasibility of implementing adaptive systems. In autonomous vehicles, techniques like reinforcement learning enable dynamic adjustment of model parameters to respond to varying weather and lighting conditions, enhancing detection robustness. Similarly, in wildlife monitoring, convolutional neural networks (CNNs) combined with transfer learning are used to classify animals under occlusions and shadows, showing how adaptive pre-processing steps like brightness correction and noise reduction can mitigate environmental effects.

In the railway context, future systems could integrate such adaptive approaches, employing techniques like domain adversarial training to handle diverse lighting conditions and weather scenarios. Additionally, infrared and thermal imaging technologies, already effective in surveillance applications, could be combined with visible light imaging to improve detection accuracy under low-light or adverse weather conditions. These adaptive measures will enable

railway computer vision systems to maintain high performance across a wide range of environmental variables.

6.2 Real-time processing and computational constraints

Real-time processing is a critical requirement in railway operations, especially when monitoring high-speed trains or busy passenger stations. However, the computational demands of computer vision models often pose significant challenges. Systems that rely on complex models, such as the CNN-based fall detection by Alawad et al. [38] and the Bayesian deep learning framework by Wu et al. [70] for bogie fault diagnosis, require high processing power to handle the large volume of data generated in real-time. Processing high-resolution images or video feeds while ensuring low latency remains a bottleneck, especially on edge devices or in mobile deployment scenarios.

Studies such as Alif and Hussain [19] emphasised the computational demands of deep learning models, even when optimised for edge devices. Their lightweight CNN, integrated with attention mechanisms, achieved high accuracy for bolt detection while maintaining computational efficiency. However, even this optimised approach struggled to process very high-resolution images in real-time, demonstrating the inherent challenges of balancing accuracy and efficiency in constrained environments. Hardware limitations also constrained the scalability of systems such as VISyR, which employs an FPGA-based setup for bolt detection at speeds up to 201 km/h but relies on specialised hardware for real-time processing [45]. Additionally, models that require substantial computational resources, such as the Mask R-CNN used for trespassing detection [39], may be challenging to deploy across extensive railway networks due to hardware costs and the need for constant upgrades. Addressing these computational constraints may require model optimisation techniques, such as pruning, quantisation, or the development of lightweight models, to achieve a balance between accuracy and efficiency. Sun et al. [31] also demonstrated a binocular vision-based bolt-loosening detection system capable of detecting faults on running trains within five minutes for an eight-car train. However, the system required high-performance GPUs, limiting its scalability for larger railway networks or higher-speed operations.

The advancements in lightweight models and hardware optimisation are promising but insufficient for the demands of real-time monitoring in high-speed railways. Future work must focus on developing more computationally efficient models, such as those employing pruning, quantisation, or lightweight architectures, to enable faster and more scalable real-time defect detection systems.

Table 6 Summary of applications of computer vision for rail engine components defect detections

Model	Dataset used	Detection task	Performance	References
Anchor-Free YOLOv8 (AFYv8)	Custom dataset of bogie parts captured from high-speed action cameras	Identification and tracking of bogie components	99.1% mAP, reliable performance between 10–40 km/h	[46]
Bayesian deep learning with wide residual network	Simulated dataset of high-speed train bogie faults	Detection of known and unknown bogie faults	Improved mAP for concurrent and unexpected faults	[70]
Multi-dimension end-to-end convolutional neural network (CNN)	Multi-sensor data from vibration signals in three orthogonal directions	Fault diagnosis of rotating devices in train bogies	Over 98% accuracy across laboratory and real-world data	[71]
Capsule network (CapsNet)	Simulated dataset from SIMPACK for seven bogie conditions	Classification of single and compound faults in train bogies	96.65% accuracy in diagnosing seven bogie conditions	[72]
Cross-correlation analysis	Experimental and simulated datasets of vertical accelerations in bogie suspensions	Fault detection in suspension dampers	Reliable fault detection through CCC analysis	[73]
Vision transformer (ViT)	Bolt-specific image dataset (145 images)	Classification of missing bolts	93% accuracy	[18]
Lightweight convolutional neural network (CNN) with integrated attention mechanism	Custom dataset of railway bolt images	Detection of missing bolts	96.43% accuracy with F1-score of 96	[19]
YOLOv3	500 images of bolts on steel bridges (augmented)	Detection of delayed fractures in bolts	AP of 89.14% with IoU threshold of 0.5	[78]
Binocular stereo vision system	Stereo images of bolts on running trains	Detection of bolt-loosening in key train components	Relative error < 1.42%; Repeatability SD < 0.03 mm	[31]
CNN with virtual coding and itti encoder	Subway vent valve images labelled by operational state	Monitoring subway vent valve positions	98.5% accuracy, F1-score of 99.6%	[79]
Nanodet-ResNet101	4,451 images of train height valve components	Detection of faults in train height valves	97% accuracy, 0.1 s processing time per image	[80]

6.3 Data availability and model generalisation

Another key challenge in railway computer vision applications is the availability of diverse, high-quality datasets for training and testing. Many models, including the bolt detection system by Hua et al. [17] and the crowd monitoring framework by Zhang et al. [42], rely on custom or domain-specific datasets. However, gathering sufficient labelled data in railway environments is difficult due to privacy concerns, limited access to certain railway areas, and the variability of track and component conditions. Gorane and Joshi [68] highlighted the limitations of annotated datasets, particularly for passenger detection, where railway-specific scenarios are often under-represented. Moreover, the development of robust computer vision systems often requires large, annotated datasets to train and validate models. Railway-specific datasets, however, are frequently limited in size, variety, and accessibility. For example, Alif et al. [18] noted that the lack of diverse annotated datasets for bolt detection restricted model generalisability, leading to potential overfitting during training. Similarly, Liu et al. [65] observed that the small size of datasets in BBK detection further compounds this issue, as the limited data does not adequately represent the full spectrum of real-world scenarios.

Generalisation remains a significant hurdle when applying models trained on specific datasets to diverse railway environments with different track designs, lighting, or operational practices. For example, models like Fault-Net, Pahwa et al. [33], and the multidimensional CNN for bogie fault diagnosis by Kou et al. [19] may face reduced accuracy in environments not represented in the training data. Variability in railway configurations, train speeds, and environmental conditions can lead to performance

degradation when models encounter unfamiliar scenarios. To overcome these challenges, future efforts must focus on collaborative data sharing between railway organisations to build comprehensive and diverse datasets. The adoption of synthetic data generation and domain adaptation techniques could also play a pivotal role in addressing dataset limitations. Additionally, transfer learning, as seen in Zhou et al. [78] for height valve fault detection, offers a promising solution by leveraging pre-trained models to improve generalisation across diverse railway scenarios.

While computer vision offers promising advancements for railway safety and maintenance, the practical deployment of these systems faces significant challenges. Environmental conditions, computational constraints, and data limitations hinder the scalability and effectiveness of current solutions. Addressing these challenges through technological innovations, such as adaptive algorithms, hardware optimisation, and diverse dataset collection, will be essential for realising the full potential of computer vision in railway applications. A summary of these challenges and proposed solutions is given in Table 7. These improvements will not only enhance the accuracy of obstacle detection and component fault diagnosis but also contribute to the safety and reliability of passenger monitoring systems in crowded stations and high-speed rail environments.

6.4 Self-supervised, semi-supervised, and generative learning for data-scarce railway inspection

A persistent challenge in railway computer vision is the scarcity of labelled defect data. Many railway faults, including

Table 7 Challenges, impact on detection, and proposed solutions in railway computer vision systems

Challenge	Impact on detection	Proposed solutions	References
Low lighting conditions	Reduced accuracy and false negatives in image-based detection tasks	Enhanced lighting setups, use of high-resolution cameras, and data augmentation techniques for low-light environments	[17, 42, 65]
Weather effects (rain, fog, snow)	Occlusions and noise in image data can lead to increased false positives	Use of infrared imaging, adaptive image processing algorithms, and synthetic data generation to simulate adverse weather scenarios	[19, 39, 68]
Motion blur in high-speed operations	Misclassification or inability to detect fine-grained defects	High-speed cameras, motion-compensated algorithms, and real-time image stabilisation techniques	[46, 74]
Limited computational resources	Delayed real-time processing and reduced throughput in high-resolution image analysis	Lightweight model architectures (e.g. Nano-det, YOLOv5-Tiny), edge computing, and model pruning	[17, 19, 80]
Lack of diverse datasets	Poor model generalisation, overfitting to specific conditions	Collaborative dataset sharing, synthetic data generation, and transfer learning from broader datasets	[17, 55, 65]
Variation in railway environments (tracks, speeds, axle loads)	Reduced adaptability and accuracy when applied to new scenarios	Domain adaptation techniques, multi-task learning, and domain adversarial training	[27, 55, 70]

cracked rails, missing fasteners, and component failures in rolling stock, are rare events that occur infrequently during normal operations. As a result, large-scale manual annotation is costly, time-consuming, and often impractical. While the majority of existing railway inspection systems rely on fully supervised learning, recent research has increasingly explored semi-supervised and self-supervised learning paradigms to mitigate these limitations.

Semi-supervised learning techniques leverage a combination of limited labelled data and abundant unlabelled imagery to improve model generalisation. Approaches such as pseudo-labelling, consistency regularisation, and teacher–student frameworks have demonstrated strong effectiveness in industrial surface defect detection. Liu et al. [52] provide a comprehensive overview of semi-supervised strategies for defect detection in industrial environments, highlighting their ability to reduce annotation requirements while maintaining competitive performance. In the railway domain, Ozdemir and Koc [53] show that pseudo-label-based semi-supervised learning can significantly enhance defect detection accuracy under sparse annotation regimes. More recently, Ozdemir et al. [83] proposed a student–teacher framework that automatically assigns high-confidence pseudo-labels to unlabelled railway defect images, effectively expanding the training dataset and improving detection robustness. Complementary studies in industrial inspection further support these findings: Liu et al. [84] emphasise the effectiveness of pseudo-labelling and consistency training for surface defect classification, while Shi et al. [85] demonstrate that semi-supervised segmentation frameworks can alleviate performance degradation caused by limited defect samples. Although railway-specific semi-supervised studies remain relatively few, these approaches are well aligned with operational railway settings, where labelled fault data are scarce but large volumes of unlabelled inspection imagery are readily available.

Beyond semi-supervised approaches, self-supervised learning has emerged as a powerful paradigm for representation learning without manual annotations. In this setting, models are pre-trained using pretext tasks such as contrastive learning, image reconstruction, or temporal consistency, before being fine-tuned for downstream inspection tasks. Self-supervised techniques have shown strong performance in industrial anomaly detection by learning robust feature representations from unlabelled data. In the railway context, Min et al. [86] employ a variational autoencoder (VAE) trained exclusively on normal rail surface images, detecting anomalies through reconstruction residuals and thereby reducing dependence on labelled defect samples. Li et al. [87] further categorise self-supervised and unsupervised anomaly detection methods as key solutions for data-scarce industrial inspection scenarios, highlighting their ability to exploit unlabelled data for representation learning

and anomaly scoring. The adoption of self-supervised learning in railway inspection therefore holds substantial potential for improving robustness across diverse environmental conditions, asset types, and sensor configurations while minimising annotation effort.

Generative learning represents another promising direction for addressing data scarcity in railway inspection. Generative models, including variational autoencoders, generative adversarial networks, and diffusion-based frameworks, can synthesise realistic defect images to augment rare fault classes and mitigate severe class imbalance. Although railway-specific generative studies remain limited, generative approaches are widely used in industrial image analysis to produce high-fidelity synthetic defect data suitable for training deep learning systems. Recent self-supervised anomaly detection research also integrates synthetic anomaly insertion during training to enhance detection performance without requiring manual labels [88]. In the railway domain, generative models offer a safe and practical means of simulating rare or hazardous fault scenarios that are difficult to capture during real operations, thereby supporting more robust training and evaluation of inspection systems.

Overall, self-supervised, semi-supervised, and generative learning techniques provide a complementary toolkit for overcoming one of the most significant barriers to large-scale deployment of computer vision in railway engineering. Future research should focus on systematically evaluating these approaches under real-world railway conditions, integrating them with multi-sensor data and Remote Condition Monitoring systems, and establishing best practices for their safe, interpretable, and reliable deployment in safety-critical inspection workflows.

6.5 Advancements in deep learning and AI models

The rapid evolution of deep learning and broader machine learning approaches has significantly transformed computer vision applications in railway engineering. Existing railway monitoring systems already play a central role in ensuring safety and operational efficiency, with technologies such as axle bearing monitoring, wheel defect detection, and track geometry assessment being well established [7]. Deep learning models now introduce new opportunities to enhance these systems by improving defect recognition accuracy, increasing automation, and reducing operational costs. Architectures such as ViTs [18] and advanced CNNs [22, 23] offer strong performance in recognising complex spatial patterns under challenging environmental conditions. The YOLO family of models [26], widely used for real-time object detection, has been adapted for railway inspection tasks, enabling rapid and accurate identification of defects on tracks and rolling stock.

Transfer learning has also become an effective strategy for railway applications, allowing pre-trained deep learning models to be adapted to domain-specific datasets with minimal additional training [89]. This approach reduces computational requirements and accelerates deployment in operational settings. Unsupervised learning, which is a subfield of machine learning, is gaining attention for railway inspection because it identifies patterns and anomalies without relying on large labelled datasets. This is particularly beneficial given the difficulty of obtaining annotated defect data in railway environments [90].

Deep learning models are increasingly integrated with multi-sensor systems to improve decision-making and robustness. By combining information from RGB cameras, thermal sensors, and LiDAR, these systems can detect a wider range of defects under diverse environmental conditions [62, 63]. Edge AI, which refers to running machine learning models directly on resource-constrained devices such as inspection robots and onboard processors, is becoming essential for real-time deployment [46, 61]. This reduces reliance on centralised processing and supports immediate response in high-speed railway environments.

Emerging research is exploring generative AI models [54] and hybrid deep learning architectures that combine transformer layers with convolutional operations [50, 91]. These models are designed to process large-scale multimodal data efficiently and hold promise for enabling fully autonomous railway monitoring and maintenance pipelines. As the railway industry continues to adopt these technologies, deep learning and AI-driven systems are expected to play an increasingly important role in improving safety, operational efficiency, and long-term sustainability.

Remote Condition Monitoring (RCM) remains a key pillar of railway asset management. Current RCM systems monitor components such as axle bearings, pantographs, wheel defects, and track geometry [7, 14, 36]. The integration of computer vision, machine learning, and Internet of Things (IoT) technologies offers a powerful extension to these systems. Tucker et al. [92] identified challenges in the wider adoption of RCM, including lack of standardisation, misaligned incentives, and concerns surrounding data ownership. Their work highlights the importance of coordinated data-sharing frameworks for unlocking the full potential of predictive maintenance.

The combination of computer vision with IoT sensors provides enhanced detection capability by merging visual data with thermal, vibration, pressure, and structural measurements [62, 63]. Deep learning models can process these multimodal inputs to detect defects such as cracks, wear, and structural deformation even at high speeds [41, 61]. This enables proactive maintenance strategies that surpass the limitations of traditional ultrasonic or eddy-current

inspection methods, which often require controlled conditions and reduced operating speeds [93].

IoT-enabled sensors embedded within railway assets can continuously monitor parameters including temperature, load, and stress [94]. When combined with computer vision systems, they provide additional diagnostic information that enhances the localisation and interpretation of faults [95]. The fusion of RGB and thermal data, demonstrated in the 3D RGB-T model by Wang et al. [96], improves defect visibility and enables detection of subsurface damage that may not appear in standard images. Real-time communication between IoT devices supports dynamic scheduling of inspections and contributes to reduced downtime [97]. Edge computing further enhances responsiveness by processing sensor data directly at the source [98].

IoT integration is also a foundation for autonomous monitoring systems, including drones and ground robots equipped with computer vision [30, 63]. These platforms can conduct inspections independently and transmit diagnostic information to centralised systems for analysis. Cloud-based platforms enable large-scale data fusion and predictive analytics for maintenance planning [94], supporting a transition towards smarter and more adaptive railway networks.

As railway systems move towards digitalisation, the combined use of deep learning, machine learning, and IoT-driven sensing will be essential for developing intelligent, self-adaptive inspection and maintenance frameworks. These advancements are expected to significantly enhance safety, operational efficiency, and sustainability across global rail networks. To move beyond a paper-by-paper listing and provide a comparative perspective, the reviewed methods can be grouped according to their underlying model architectures and deployment characteristics. As summarised in Table 3, one-stage detectors such as the YOLO family are frequently preferred for real-time railway inspection tasks due to their low latency and computational efficiency, making them suitable for high speed and onboard applications. In contrast, segmentation-based architectures such as Mask R-CNN are commonly adopted when precise defect localisation and pixel-level delineation are required, for example in crack analysis or fine-grained surface defect assessment. Transformer-based models, while computationally more demanding, offer advantages in capturing long-range spatial dependencies and show promise for complex inspection scenarios involving large-scale or cluttered railway scenes.

6.6 Real-time and autonomous railway inspection systems

The advancement of real-time and autonomous railway inspection systems marks a significant leap forward in pursuing safer, more efficient railway operations. While numerous monitoring systems are already in place, such as inspection

trains for track geometry [36] and onboard axle position monitoring via stereo computer vision systems [99], computer vision offers a pathway to more efficient, scalable, and cost-effective solutions.

Real-time inspection systems leverage high-speed cameras, computer vision algorithms, and edge computing to process large volumes of visual and sensor data as it is collected [30, 61]. These systems are critical for high-speed railways, where rapid defect identification and response can prevent catastrophic failures [41]. For instance, real-time detection of rail surface cracks or misaligned fasteners can allow for immediate corrective actions, significantly reducing downtime and safety risks. Edge computing plays a pivotal role by enabling local data processing near the inspection point, reducing latency, and ensuring quick decisions [61]. This capability is especially vital for detecting dynamic faults, such as wheel defects or suspension issues, in moving trains. Furthermore, real-time systems are increasingly being integrated with centralised cloud platforms, enabling railway operators to monitor their entire network remotely and take proactive measures based on live updates [100].

Autonomous inspection systems, such as drones and mobile robots, further enhance railway maintenance by reducing dependency on human labour and allowing for continuous, 24/7 operations [63]. Equipped with computer vision and AI capabilities, drones can inspect tracks, overhead lines, and bridges, while mobile robots equipped with multimodal sensors can assess components like ballast, fasteners, and sleepers at ground level [43, 63].

Field-Programmable Gate Arrays (FPGAs) are becoming an increasingly popular choice for real-time railway monitoring systems due to their ability to deliver high performance with low latency [45]. Unlike traditional processors, FPGAs offer customisable hardware configurations, allowing parallel processing of high-resolution images and video streams, which is critical for real-time applications. Systems like VISyR, which use FPGA-based architectures for bolt detection at speeds of up to 201 km/h, exemplify the potential of these devices to handle the computational demands of high-speed railway operations [45]. The adaptability of FPGAs also makes them suitable for integrating advanced computer vision models, such as convolutional neural networks (CNNs), directly onto hardware, further reducing processing time [101]. While FPGAs offer significant advantages, their specialised nature and higher initial costs can limit widespread adoption. However, ongoing advancements in FPGA technology, including the development of lower-cost options and integration with edge computing platforms, are likely to make these systems more accessible for large-scale railway applications [102].

The integration of real-time and autonomous systems with predictive maintenance frameworks further amplifies their utility. By continuously monitoring the condition of

railway assets and analysing historical data trends, these systems can predict potential failures and recommend timely maintenance. For example, a real-time system might detect unusual wear on a rail joint, triggering an autonomous robot to perform a closer inspection and send detailed diagnostics to maintenance teams [103]. Outputs from the monitoring system could, in the future, feed into automated maintenance planning tools, such as the automated fleet maintenance planning system proposed by Louadah et al. [104].

Despite their potential, real-time and autonomous systems face challenges in deployment. Ensuring consistent performance in varying environmental conditions, such as extreme weather or low lighting, remains a technical hurdle [35]. Autonomous systems also require robust navigation algorithms to operate safely and effectively in complex railway environments [32]. Future advancements may focus on integrating these systems with IoT networks and advanced AI models, such as reinforcement learning, to enhance their adaptability and decision-making capabilities [32]. As technology evolves, the railway industry is likely to see a shift towards fully automated inspection and maintenance workflows, creating smarter, safer, and more efficient railway networks.

In addition to camera-based systems and edge computing, LiDAR technology is playing an increasingly important role in railway inspection, offering high-resolution 3D mapping capabilities. Recent applications of LiDAR in railway environments are discussed below.

6.7 LiDAR applications in railway inspection

LiDAR technology has emerged as a valuable tool in railway inspection by providing high-resolution 3D point clouds that enable accurate measurement of ballast geometry, obstacle detection, and infrastructure monitoring. Zhang et al. [44] proposed a camera and LiDAR data fusion method for railway object detection, demonstrating significant improvements in identifying track components and obstacles under diverse environmental conditions. LiDAR sensors are also increasingly integrated with UAV platforms for aerial surveys of railway corridors, offering detailed topographical mapping and vegetation monitoring. Furthermore, studies such as Gbadamosi et al. [101] highlighted the use of LiDAR-enhanced IoT systems for predictive maintenance in the UK rail industry, showcasing its potential for real-time track and asset monitoring. These advancements underscore LiDAR's critical role in enhancing the robustness and accuracy of railway inspection systems, particularly when combined with computer vision and sensor fusion techniques.

The dataset summary provided in Table 8 highlights the strong dependence of current computer vision systems on custom, domain-specific datasets, many of which are limited

Table 8 Summary of datasets used in representative studies discussed in Sect. 6

Study	Application	Data modality and platform	Dataset characteristics and limitations
Alawad et al. [38, 69]	Passenger safety and fall detection in stations	CCTV video from indoor station environments	Custom surveillance datasets with strong dependence on camera placement and lighting. Limited representation of rare events such as falls and near misses
Zhang et al. [40]	Near-miss detection at level crossings	Fixed roadside surveillance cameras	Custom crossing video dataset. Performance is sensitive to weather variability, shadows, and background noise. Limited scalability across different crossing layouts
Zaman et al. [39]	Rail trespassing detection	Trackside surveillance cameras	Custom video dataset containing rare trespass events, leading to severe class imbalance and limited behavioural diversity
Gorane and Joshi [68]	Passenger detection and hazard avoidance	Station/platform imagery CCTV	Custom datasets focused on hazardous passenger regions. Under-representation of occluded scenes and crowded scenarios reduces robustness
Zhang et al. [42]	Crowd behaviour monitoring	CCTV footage and simulated crowd scenes	Combination of real and synthetic scenes. Rare behaviours such as falling or fighting produce imbalanced datasets with limited generalisation
Hua et al. [17]	Bolt condition monitoring	On-track robotic inspection platform (RGB images)	Custom bolt dataset with multiple defect types. Limited environmental diversity affects robustness under different lighting and track conditions
Alif and Hussain [18, 19]	Missing bolt detection on rolling stock/track	High-resolution RGB images	Custom bolt datasets with small sample sizes and constrained viewpoints. Class imbalance between normal and defective cases is noted
Liu et al. [65]	BBK inspection on freight wagons	High-speed video sequences	Dataset of axle box lateral retaining keys with limited sequences. Captured in controlled environments with low variation in lighting and clutter
Sun et al. [31]	Bolt-loosening detection on running trains	Stereo image pairs captured onboard	Stereo vision dataset acquired in service. Sensitive to illumination extremes and requires consistent camera mounting
Marino et al. (VISyR) [45]	High-speed bolt detection	Monochrome high-speed camera data	Dataset captured at speeds up to 201 km/h. Motion blur and occlusions remain challenging. No publicly available benchmark
Kou et al. [71]	Fault diagnosis of rotating bogie devices	Multi-channel vibration sensor signals	Hybrid laboratory and in-service vibration dataset. Limited coverage of rare compound faults and not publicly accessible
Zhang et al. [55]	Absent sleeper support detection	Simulated and measured displacement signals	Dataset combining FEM simulations and field measurements. Accuracy depends on the fidelity of simulation-to-reality alignment

in scale, diversity, and environmental variability. These constraints directly influence the robustness and generalisability of real-time and autonomous inspection solutions. As research continues to advance, integrating richer multimodal datasets, improving data-sharing practices across the rail industry, and adopting synthetic or semi-supervised learning techniques will be essential for developing more adaptable and reliable monitoring systems. By combining enhanced data resources with high-speed imaging, edge computing, and autonomous platforms, future inspection frameworks are positioned to deliver faster, safer, and more scalable railway maintenance across complex operational environments.

7 Conclusion

This review has examined the growing role of computer vision in modern railway engineering, highlighting its potential to transform inspection, maintenance, and safety monitoring. Across track infrastructure, rolling stock systems, and passenger environments, computer vision and deep learning models offer substantial improvements in accuracy, consistency, and scalability compared with traditional manual inspection practices. These advancements directly address long-standing operational challenges such as real-time defect detection, reduction of human error, and the optimisation of maintenance workflows. By automating labour-intensive visual assessments, computer vision provides a foundation for more reliable and efficient railway operations. The review followed a structured methodology drawing on a wide range of peer-reviewed studies, enabling balanced coverage of both technical progress and practical system limitations.

The findings show that challenges differ across the three principal application domains. For rolling stock monitoring, issues arise from the need to process high speed and high-volume visual data, maintain robustness under variable lighting and environmental conditions, and detect diverse mechanical faults. Advances such as Capsule Networks for compound fault analysis and Bayesian deep learning for uncertainty estimation illustrate promising directions for diagnosing axle, suspension, and wheel-related defects. In track monitoring, limitations primarily relate to dataset diversity, environmental variability, and real-time computational requirements. Deep learning models such as Mask R-CNN, YOLO, and Deeplabv3+ have shown strong performance in detecting cracks, geometric deviations, and surface wear, while the integration of RGB, LiDAR, and thermal imaging is beginning to overcome the weaknesses of single-modality systems.

For safety monitoring, the key challenge lies in recognising dynamic, unpredictable events such as trespassing or overcrowding in real-time. Although models such as DFF-Net and Mask R-CNN have achieved high accuracy

in controlled settings, further robustness is required before widespread deployment in diverse, real-world passenger environments can be achieved. Among the three domains, track inspection appears closest to large-scale deployment due to the maturity of existing models and the emergence of commercial and pilot systems. Demonstrations of YOLO-based defect detection and UAV-assisted monitoring already show practical feasibility. Rolling stock and safety monitoring show comparable potential but require additional research to address environmental variability, high-speed imaging constraints, and generalisation across different operational contexts.

A recurring theme in the reviewed literature is the benefit of integrating computer vision with existing RCM systems. Combining visual data with ultrasonic measurements, thermal imaging, and vibration signals provides a richer representation of asset condition and reduces the limitations of any single sensing modality. Such multi-sensor fusion supports more accurate decision-making, reduces false alarms, and enhances predictive maintenance capabilities. Model interpretability has also emerged as an important enabler for practical adoption. While deep learning models offer high accuracy, their opaque decision processes remain a barrier in safety-critical contexts. Techniques such as Grad-CAM help visualise model attention and provide engineers with insight into model reasoning, improving transparency, accountability, and regulatory readiness. Strengthening interpretability will therefore be essential for building trust in AI-driven inspection systems.

Taken together, the findings of this review highlight computer vision as a central technology in the evolution of intelligent and resilient railway systems. Continued progress in deep learning architectures, real-time processing, dataset diversity, and multi-sensor integration will be critical for enabling scalable deployment. As these technologies mature and transition from research prototypes to operational systems, they hold considerable potential to deliver safer, more efficient, and more sustainable railway networks. In combination with predictive maintenance frameworks and modern RCM platforms, computer vision will play a key role in shaping the next generation of automated, data-driven railway inspection and monitoring.

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